# The Thiele Machine

A Complete Mathematical Specification

Derived from the Fully Machine-Checked Coq Corpus

Devon Thiele

April 2026

**Abstract**

This is the complete mathematical specification of the Thiele Machine: a formal model of computation where structural information carries an explicit, conserved cost $\mu$. Everything here is built from the machine-checked Coq corpus. You can reconstruct the machine from what's in this document alone.

Coq establishes that conclusions follow from axioms. It does *not* establish that axioms model physical reality. That distinction matters and I keep it explicit throughout. Every result is tagged as one of three types: **(S)** structural theorems about the machine as a mathematical object, **(C)** conditional derivations where a physics conclusion follows from a stated axiom, and **(R)** consistency relations that check internal coherence but are not independent predictions.

The document covers: the full state space and operational semantics; No Free Insight; $\mu$-ledger necessity and minimality; computational universality and Turing strictness; gauge symmetry and Noether conservation; conditional connections to quantum mechanics (Born rule, Tsirelson bound, no-cloning, unitarity, complex amplitudes, Schrödinger equation) under named physical axioms; spacetime emergence and the Landauer bridge; the Thiele Manifold and self-reference tower; and falsifiable predictions with concrete failure modes.

# Contents

# Chapter 1

# Introduction

The Thiele Machine makes the cost of structure explicit. A Turing Machine treats all state transitions as cost-equivalent. The Thiele Machine assigns a specific cost $\mu$ to every operation that extracts or asserts structural information. That's the design. Here is where it goes.

The central claim, proven formally: *you cannot gain structural insight about a system without paying $\mu$-cost.* Combined with explicit physical axioms (superposition, linearity, information conservation), that constraint produces structural results that parallel quantum mechanics, thermodynamics, and emergent spacetime. Each result's status — whether it is a structural theorem **(S)**, a conditional derivation **(C)**, or a consistency relation **(R)** — is marked directly. Nothing is hidden.

Eight parts:

1. **Foundations**: State space, instruction set, operational semantics.
2. **Core Theorems**: No Free Insight, $\mu$-ledger necessity, $\mu$-Chaitin, cost = complexity.
3. **Computation**: Universality, strict subsumption, confluence, halting.
4. **Symmetry & Conservation**: $\mu$-conservation, Noether/gauge, receipts.
5. **Quantum Mechanics**: Born rule, Tsirelson, no-cloning, unitarity, complex amplitudes, Schrödinger, Planck.
6. **Emergent Structure**: Spacetime metric, causal cones, information causality, Landauer bridge.
7. **Meta-Theory**: Self-reference, Thiele Manifold, Genesis, three-layer isomorphism, Curry–Howard–Thiele.
8. **Predictions**: Falsifiable divergences from standard QM.

## 1.1 Epistemological Framework

Formal verification (Coq) establishes that conclusions follow logically from axioms. It does **not** establish that axioms model physical reality. This distinction is critical for evaluating the physics claims in Parts V–VIII.

Every result in this specification falls into one of three categories:

**(S) Structural** Theorems about the Thiele Machine as a mathematical object. These are unconditionally true given the definitions. *Examples*: $\mu$-conservation, gauge invariance, confluence, halting undecidability, strict Turing subsumption.

**(C) Conditional** Theorems of the form "if axiom $X$ holds, then physics result $Y$ follows." The logical inference is verified; whether axiom $X$ holds in nature is an empirical question. *Examples*: Born rule uniqueness (conditional on linearity), complex necessity (conditional on 2D amplitudes), unitarity (conditional on information conservation).

**(R) Consistency** Algebraic identities or definitional equivalences that verify internal coherence but do not constitute independent predictions. *Examples*: Planck consistency relation, Tsirelson cost definition.

Each major theorem in Parts V–VIII is annotated with its category. A theorem marked **(C)** or **(R)** is not diminished. Conditional derivations and consistency checks are valuable scientific tools. But the reader should not mistake them for derivations from first principles without stated assumptions. See Appendix B for the complete classification table.

# Chapter 2

# The State Space

## 2.1 Primitives

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. Let $\mathbb{B} = \{0, 1\}$. Let $\Sigma$ be a finite alphabet.

## 2.2 Regions and Modules

**Definition 2.1** (Region)**.** A **Region** $R$ is a finite subset of $\mathbb{N}$, canonically represented as a deduplicated list preserving first-occurrence order: $\mathrm{norm}(L) = \mathrm{nodup}(L)$ removes duplicate elements while preserving the order of their first appearance. (The Python VM applies a stricter canonical form via `sorted(set(region))`; the Coq specification uses `nodup Nat.eq_dec` which does not sort.)

**Definition 2.2** (Module)**.** A **Module** $M = (R_M, A_M)$ consists of a normalized region $R_M \subset \mathbb{N}$ and a set of axioms $A_M \subset \Sigma^*$.

## 2.3 The Partition Graph

**Definition 2.3** (Partition Graph)**.** A partition graph $G = (N_{id}, \mathcal{T})$ where $N_{id} \in \mathbb{N}$ is the next available module ID and $\mathcal{T} : \{0, \dots, N_{id} - 1\} \rightharpoonup \mathcal{M}$ is a finite partial function from IDs to modules.

**Axiom 2.1** (Well-Formedness)**.** *$G$ is well-formed iff $\mathrm{dom}(\mathcal{T}) \subseteq \{0, \dots, N_{id} - 1\}$.*

## 2.4 Machine State

**Definition 2.4** (VM State)**.** The complete state is a 12-tuple:

$$S = (G, C, R, \mathit{Mem}, PC, \mu, T_\mu, \mathit{err}, \ell, m, w, \mathit{cert})$$

where $G \in \mathcal{G}$ is the partition graph, $C \in \mathbb{N}^4$ comprises the CSRs (certificate address, status, error code, heap base), $R \in \mathbb{N}^{32}$ is the register file, $\mathit{Mem} \in \mathbb{N}^{65536}$ is main memory, $PC \in \mathbb{N}$ is the program counter, $\mu \in \mathbb{N}$ is the $\mu$-ledger, $T_\mu \in \mathbb{N}^{16}$ is the flattened $4 \times 4$ $\mu$-tensor (row-major), $\mathit{err} \in \mathbb{B}$ is the global error flag (latching), $\ell \in \mathbb{N}$ is the logic-engine accumulator, $m \in \mathbb{N}$ is the machine status register ($0 =$ Turing mode, $1 =$ Thiele mode), $w \in \mathbb{N}^8$ is the WitnessCount record (`vm_witness`): eight natural-number counters `wc_same_ab` and `wc_diff_ab` for each setting pair $(a, b) \in \{00, 01, 10, 11\}$, and $\mathit{cert} \in \mathbb{B}$ is the certified flag (`vm_certified`).

# Chapter 3

# Instruction Set and Cost Model

## 3.1 Instruction Categories

The instruction set $\mathcal{I}$ is organized into six families. Every instruction $i$ carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$.

### 3.1.1 Structural Operations

These modify the partition graph $G$:

- `PNEW`$(R, \Delta\mu)$: Create module with region $R$.
- `PSPLIT`$(m, L, R, \Delta\mu)$: Split module $m$ into disjoint regions.
- `PMERGE`$(m_1, m_2, \Delta\mu)$: Merge two modules.
- `PDISCOVER`$(m, E, \Delta\mu)$: Discovery accounting for module $m$. The abstract specification attaches evidence $E$ (list of axioms) to the module; the current hardware-aligned step advances PC and charges $\Delta\mu$ without mutating the graph.
- `MDLACC`$(m, \Delta\mu)$: Minimum-description-length accumulate on module $m$.

### 3.1.2 Logical Operations

These interact with information content:

- `LASSERT`$(\mathit{freg}, \mathit{creg}, \mathit{kind}, \mathit{flen}, \mathit{cost})$: Assert a formula read from `vm_mem`[`R[freg]`] (encoded-unit count *flen*, eight concrete bits per unit) against a witness package at `vm_mem`[`R[creg]`]. In SAT mode the package contains one satisfying assignment and one falsifying assignment, stored back-to-back; $\Delta\mu = \mathit{flen} \times 8 + S(\mathit{cost}) \geq 1$.
- `LJOIN`$(\mathit{c1reg}, \mathit{c2reg}, \Delta\mu)$: Join certificates from two memory-resident strings addressed by registers.
- `REVEAL`$(m, n, \mathit{cert}, \Delta\mu)$: Reveal $n$ bits of structure from $m$; cost is $n + S(\Delta\mu)$.
- `EMIT`$(m, p, \Delta\mu)$: Emit payload $p$ from module $m$; cost is $|p|_{\text{bits}} + S(\Delta\mu)$.

### 3.1.3 Computational Operations (Reversible ALU)

Reversible register/memory operations:

- `XFER`$(\mathit{dst}, \mathit{src}, \Delta\mu)$: Copy register *src* to *dst*.

- `XOR_LOAD`$(dst, addr, \Delta\mu)$: Load memory word at $addr$.
- `XOR_ADD`$(dst, src, \Delta\mu)$: $dst \leftarrow dst \oplus src$.
- `XOR_SWAP`$(a, b, \Delta\mu)$: Swap registers $a$ and $b$.
- `XOR_RANK`$(dst, src, \Delta\mu)$: $dst \leftarrow \text{popcount}(src)$.

#### 3.1.4 Standard ALU and Control Flow

- `LOAD_IMM`$(dst, imm, \Delta\mu)$: Load 32-bit immediate into register.
- `LOAD`$(dst, addr, \Delta\mu)$: Load from main memory.
- `STORE`$(addr, src, \Delta\mu)$: Store to main memory.
- `ADD`$(dst, r_1, r_2, \Delta\mu)$: $dst \leftarrow r_1 + r_2 \pmod{2^{64}}$.
- `SUB`$(dst, r_1, r_2, \Delta\mu)$: $dst \leftarrow r_1 - r_2 \pmod{2^{64}}$.
- `JUMP`$(target, \Delta\mu)$: Unconditional jump.
- `JNEZ`$(r, target, \Delta\mu)$: Jump if $R[r] \neq 0$.
- `CALL`$(target, \Delta\mu)$: Push return address, jump to target.
- `RET`$(\Delta\mu)$: Pop return address, jump back.

#### 3.1.5 Auxiliary Operations

- `CHSH_TRIAL`$(x, y, a, b, \Delta\mu)$: Run a CHSH correlation trial; $x, y, a, b \in \{0, 1\}$ required.
- `CHECKPOINT`$(label, \Delta\mu)$: Execution checkpoint (charges $\mu$).
- `READ_PORT`$(dst, ch, val, bits, \Delta\mu)$: Read from I/O channel (cert-setter).
- `WRITE_PORT`$(ch, src, \Delta\mu)$: Write to I/O channel.
- `HEAP_LOAD`$(dst, addr, \Delta\mu)$: Load from heap-relative address.
- `HEAP_STORE`$(addr, src, \Delta\mu)$: Store to heap-relative address.
- `HALT`$(\Delta\mu)$: Halt execution.

#### 3.1.6 Categorical Morphism Operations

These 7 opcodes (0x27–0x2D) implement the categorical layer over partition modules. See Chapter 30 for laws and proofs.

- `MORPH`$(dst, src, tgt, idx, \Delta\mu)$: Create morphism $src \to tgt$; write morphism ID to `regs`$[dst]$.
- `COMPOSE`$(dst, f, g, \Delta\mu)$: Compose $f; g$ when $f.\text{tgt} = g.\text{src}$; errors on type mismatch.
- `MORPH_ID`$(dst, m, \Delta\mu)$: Create identity morphism $\text{id}_m$ (diagonal relation).
- `MORPH_DELETE`$(f, \Delta\mu)$: Remove morphism $f$; cascades when modules are split/merged.
- `MORPH_ASSERT`$(f, prop, cert, \Delta\mu)$: **Cert-setter.** Charges $S(\Delta\mu) \geq 1$ unconditionally. Even failed attempts (morphism does not exist) incur cost 1. This is the strongest form of the NoFI law: you cannot attempt to certify a relational claim for free.

- `MORPH_TENSOR`$(dst, f, g, \Delta\mu)$: Tensor product $f \otimes g$ (requires union modules to exist).
- `MORPH_GET`$(dst, f, sel, \Delta\mu)$: Read morphism metadata: selector $0$ = src, $1$ = tgt, $2$ = coupling length, $3$ = is_identity.

The active implementation exposes a 46-opcode ISA (39 original + 7 categorical morphism opcodes). This condensed mathematical specification presents the core instruction families relevant to the kernel semantics and cost model. Every instruction carries an explicit $\Delta\mu$ parameter; the function `instruction_cost` extracts it.

## 3.2 Cost Model

Every instruction $i$ carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$. The function `instruction_cost`$(i)$ extracts this parameter. The cost is applied via `apply_cost`$(s, i) = s.\mu +$ `instruction_cost`$(i)$. Since $\Delta\mu_i : \mathbb{N}$, the $\mu$-ledger is monotonically non-decreasing by construction.

The mandatory positive-cost class is: `CERTIFY`, `MORPH_ASSERT`, and `LJOIN` (each charges $S(\Delta\mu) \geq 1$); `EMIT` ($|payload|_{\text{bits}} + S(\Delta\mu)$); `REVEAL` and `READ_PORT` ($bits + S(\Delta\mu)$); and `LASSERT` ($flen \times 8 + S(\Delta\mu)$). These cannot be zero-cost even with $\Delta\mu = 0$. All other instructions charge exactly their encoded $\Delta\mu$, which is 0 in typical programs.

# Chapter 4

# Operational Semantics

## 4.1 Transition Function

The transition $\delta : \mathcal{S} \times \mathcal{I} \to \mathcal{S}$ is defined by:

1. $PC' = PC + 1$ for non-control-flow instructions; JUMP, JNEZ, CALL, and RET set PC to the target address; HALT stops execution.

2. $\mu' = \mu + \Delta\mu_i$.

3. $err' = err \vee \text{fail}(S, i)$.

## 4.2 Traces and Execution

**Definition 4.1** (Trace)**.** A trace $\tau = [i_0, \ldots, i_k] \in \mathcal{I}^*$ is a finite sequence of instructions.

**Definition 4.2** (Execution)**.** $\text{Run}([], S) = S; \quad \text{Run}(i :: \tau', S) = \text{Run}(\tau', \delta(S, i))$.

## 4.3 Selected Instruction Semantics

**PNEW**: Normalize region; if already present, idempotent; else add $(R_{\text{norm}}, \emptyset)$ to $G$.

**REVEAL**: Update CSRs with checksum of certificate *cert*. Cost: $\mathit{bits} + S(\Delta\mu)$.

**LASSERT**: Read formula from `vm_mem`[`R[formula_addr_reg]`] (encoded-unit count `formula_len`, eight concrete bits per unit) and witness package from `vm_mem`[`R[cert_addr_reg]`]. Verify using `cert_kind` (false = LRAT/UNSAT, true = satisfying model plus falsifying countermodel). If valid, the constraint is certified; if invalid, set $err \leftarrow$ true. Note: in the current hardware-aligned semantics, LASSERT validates the constraint but does not directly mutate the partition graph's axiom set — cert-channel activation happens through the subsequent CERTIFY or MORPH_ASSERT. Cost: `formula_len` $\times 8 + S($`mu_delta`$)$.

# Chapter 5

# No Free Insight

The central theorem: structural insight is never free.

## 5.1 Definitions

**Definition 5.1** (Receipt Predicate)**.** $P : \mathcal{O} \to \mathbb{B}$ is a computable function on observation histories.

**Definition 5.2** (Strength Ordering)**.** $P_1 \le P_2$ iff $\forall o.\ P_1(o) = 1 \implies P_2(o) = 1$. $P_1 < P_2$ iff $P_1 \le P_2$ and $\exists o.\ P_2(o) = 1 \wedge P_1(o) = 0$.

**Definition 5.3** (Certification)**.** Trace $\tau$ **certifies** $P$ iff execution succeeds ($err = 0$), the supra-certificate flag is set, and the output satisfies $P$.

**Definition 5.4** (Structure Addition)**.** HasStructureAddition($\tau$) is true iff $\tau$ contains a `REVEAL`, `LASSERT`, or equivalent operation.

## 5.2 The Theorem

**Theorem 5.1** (No Free Insight (NoFreeInsight_Theorem.v))**.** *Let* $P_{\text{strong}} < P_{\text{weak}}$. *If a trace* $\tau$ *starting from a clean state* $s_0$ *certifies* $P_{\text{strong}}$*, then:*

$$\text{Certified}(\tau, P_{\text{strong}}) \implies \text{HasStructureAddition}(\tau)$$

*Proof Sketch.* The proof is parametric: `NoFreeInsight_Theorem.v` proves the theorem for any system satisfying the `NO_FREE_INSIGHT_SYSTEM` module type interface. The interface requires four contract obligations (not Coq `Axiom` declarations—they are proved separately for the kernel instantiation): (1) $\mu$-monotonicity, (2) a certification specification linking `certifies` to execution, (3) a no-free-insight contract stating that strict strengthening requires structure addition, and (4) that the system has a notion of clean start. The theorem file itself contains zero `Axiom`, zero `Admitted`. The contract obligations are discharged by the kernel instantiation. □

**Corollary 5.2** (Cost of Insight)**.** *Gaining insight (strengthening the predicate) implies* $\Delta\mu > 0$.

## 5.3 $\mu$-Ledger Necessity

**Theorem 5.3** ($\mu$-Ledger Necessity and Complete Independence Classification (NecessityOfMuLedger.v, kernel/NecessityAbstract.v) — **(S)**)**.** *Five independence results are formally proven, zero Admitted:*

1. $\mu \perp (\mathit{mem}, \mathit{regs}, \mathit{pc})$*: no strict-classical oracle recovers* $\mu$ `turing_ram_mu_necessity`

2. $\texttt{certified} \perp (\texttt{mem}, \texttt{regs}, \texttt{pc})$: *no strict-classical oracle recovers certification* `turing_ram_cert_necessity`

3. $\texttt{certified} \perp (\texttt{mem}, \texttt{regs}, \texttt{pc}, \mu)$: *knowing $\mu$ does not determine certification* `cost_model_cert_necessity`

4. $\mu \perp (\texttt{mem}, \texttt{regs}, \texttt{pc}, \texttt{certified})$: *knowing certification does not determine $\mu$* `cert_model_mu_necessity`

5. $\texttt{vm\_graph} \perp (\texttt{mem}, \texttt{regs}, \texttt{pc}, \mu, \texttt{certified})$: *the categorical layer is independent of the full $\mu$-ledger shadow* `graph_not_recoverable_from_P_full`

***Three-component independence*** *(`thiele_state_three_component_independence`): $\mu$, `vm_certified`, and `vm_graph` are the three provably independent non-classical state components. Each is irrecoverable from any combination of the others plus $(\texttt{mem}, \texttt{regs}, \texttt{pc})$.*

***Minimality*** *(`P_full_is_minimal_complete_extension`): $(\texttt{mem}, \texttt{regs}, \texttt{pc}, \mu, \texttt{certified})$ is the minimal complete extension of classical state for $\mu$ and certification. Graph structure is a third independent axis.*

***Universality*** *(`mu_ledger_necessity_universal`): The separation holds from* every *VMState, not only from crafted zero-state witnesses. From any reachable state, `CERTIFY 0` and `PNEW [] 0` produce identical strict classical shadows but different $\mu$ and certification.*

***Trace closure*** *(`po1_trace_necessity`): For any common prefix program of any length, the two witness extensions maintain equal strict shadows but strictly different $\mu$. No accumulation of classical computation closes the gap.*

*Proof Sketch.* Each separation uses explicit witnesses where equal projected inputs force equal oracle outputs, while the witness states require unequal outputs—a contradiction. Strict shadow equality: both `CERTIFY 0` and `PNEW [] 0` advance `pc` by 1 and preserve `mem`/`regs`. $\mu$ divergence: CERTIFY charges $S(0) = 1$; PNEW charges 0. Graph separation uses `categorical_separation` from `PartitionSeparation.v`: two states with identical $P_{\text{full}}$ but different `pg_morphisms`. Universality follows because `vm_apply_certify_strict_shadow` and `vm_apply_pnew_strict_shadow` hold for arbitrary source states. □

**Theorem 5.4** (Latent $\mu$: Cost is Intrinsic to Computation (NecessityOfMuLedger.v §7) — **(S)**). *The $\mu$-cost accumulated by any computation is a property of the instruction sequence, not of any tracking system. Four results, all zero Admitted:*

1. `shadow_mu_is_computation_intrinsic`: *For any program `prog` and any starting state $s$:*

$$(\texttt{run\_instrs}\ s\ \texttt{prog}).\mu = s.\mu + \texttt{trace\_total\_cost}(\texttt{prog})$$

   *The increment is determined entirely by the instruction sequence—independent of vm_graph, vm_regs, vm_mem, vm_certified, and the starting $\mu$.*

2. `shadow_mu_delta_universal`: *Two machines running the same program from any two starting states accumulate the same additional $\mu$. The cost is written in the program before execution begins.*

3. `shadow_mu_unique_accounting`: *Any instruction-consistent accounting system $M$ satisfying $M(s) = 0$ at the starting state assigns exactly `trace_total_cost` to the result. Replace the Thiele Machine's $\mu$-counter with any other honest accounting and you get the same number.*

4. `shadow_mu_inevitable`: *Any program containing an instruction with positive declared cost accumulates strictly positive $\mu$ from any starting state. There is no free computation.*

***Interpretation.*** *A Turing machine executing a program pays exactly `trace_total_cost` in latent $\mu$ on every execution. The cost was always there—forced by the instruction sequence—but the Turing machine had no mechanism to record or expose it. The $\mu$-ledger does not create this cost. It reveals what every computation was already paying.*

## 5.4 Partition Refinement and No Free Insight

**Theorem 5.5** (Partition Ops Are Free (kernel/PartitionRefinementNoFI.v) — **(S)**)**.** *PNEW, PSPLIT, and PMERGE are not cert-setters and can carry $\Delta\mu = 0$:*

- `partition_structural_ops_not_cert_setters`*: None of the three partition operations touch* `csr_cert_addr` *or* `vm_certified`*.*
- `partition_structural_ops_can_be_free`*: All three can execute with $\Delta\mu = 0$.*
- `partition_structural_trace_cannot_certify`*: A trace containing only PNEW, PSPLIT, and PMERGE (all at $\Delta\mu = 0$) cannot reach a certified state from an uncertified start.*

*Building partition structure costs nothing by itself. Certifying structural claims about that partition is what costs.*

**Theorem 5.6** (Partition Free But Certification Non-Free (kernel/PartitionRefinementNoFI.v) — **(S)**)**.** `partition_free_but_certification_nonfree`*: There exist traces where all partition-structural instructions carry $\Delta\mu = 0$ (so $\mu_{\text{final}} = \mu_{\text{init}} = 0$) yet the trace ends with a partition graph that a subsequent MORPH_ASSERT or CERTIFY can certify — but only at $\Delta\mu \geq 1$. Partition structure can be built at zero $\mu$-cost; certified partition knowledge cannot be acquired for free. Zero Admitted.*

**Theorem 5.7** (Partition Refinement Non-Free (kernel/PartitionRefinementNoFI.v) — **(S)**)**.** `partition_refinement_nonfree`*: Any trace that starts with no cert evidence and ends with a certified claim about a partition structure paid at least 1 in $\mu$. This is the NoFI law specialized to partition refinement. Zero Admitted.*

# Chapter 6

# The $\mu$-Chaitin Theorem

A Chaitin-style incompleteness theorem denominated in $\mu$-currency.

**Theorem 6.1** ($\mu$-Chaitin (MuChaitinTheory_Theorem.v))**.** *For any formal theory system $T$ satisfying the $\mu$-Chaitin interface, the number of bits $k$ the theory can certify is bounded by:*

$$k \leq |T| + c$$

*where $|T|$ is the description length of the theory and $c$ is a fixed overhead constant. The proof chain:* $k \leq \text{payload} \leq \mu\textit{-info} \leq \text{budget} \leq |T| + c$.

# Chapter 7

# Cost Equals Kolmogorov Complexity

**Theorem 7.1** ($\mu$-Bits = Prefix-Free Complexity (CostIsComplexity.v — reference-only))**.** *Let* $K(\text{spec})$ *denote the prefix-free Kolmogorov complexity of specification* spec*. Then:*

$$\mu(\text{spec}) = K(\text{spec})$$

*Specifically,* $\mu(\text{spec}) = |\text{spec}| + 1$ *(the terminating bit), which equals the minimum description length over all prefix-free programs producing* spec*.*

*Proof.* The canonical compiler maps $\text{spec} \mapsto \text{spec} \| [\text{true}]$. Any producing program has the form $p = \text{spec} \| [\text{true}]$ with $|p| = |\text{spec}| + 1$. The compiler achieves the minimum, and $\mu$ counts exactly these bits. $\square$

# Chapter 8

# No Free Lunch and No Arbitrage

## 8.1 No Free Lunch (Ghosts Are Impossible)

**Theorem 8.1** (No Free Lunch (NoFreeLunch.v — reference-only)). *A "ghost" is defined as two distinct propositions $p \neq q$ represented by the same physical state. Ghosts are impossible:*

$$\forall\, p \neq q,\ \nexists\, s \text{ s.t. } s \text{ faithfully represents both } p \text{ and } q$$

*Information cannot exist without physical distinction.*

## 8.2 No Arbitrage Implies Potential Function

**Theorem 8.2** (Potential from No-Arbitrage (NoArbitrage.v — reference-only)). *Let $w : \mathcal{I}^* \to \mathbb{Z}$ be a cost function satisfying:*

1. ***Additivity**: $w([]) = 0$ and $w(\tau_1 \cdot \tau_2) = w(\tau_1) + w(\tau_2)$.*
2. ***No-Arbitrage**: For every closed cycle $\tau$ (returning to the starting state), $w(\tau) \geq 0$.*

*Then there exists a potential function $\phi : \mathcal{S} \to \mathbb{Z}$ such that:*

$$w(\tau) \geq \phi(\text{apply}(\tau, s)) - \phi(s)$$

*for every state $s$ and trace $\tau$.*

**Remark 8.1.** This is structurally analogous to the Second Law of Thermodynamics: consistent cost accounting implies a potential function bounding all transitions. The concrete Coq model is a toy (increment/decrement on nat with asymmetric costs). No temperature, entropy, or thermodynamic system appears in the formalization. The connection to thermodynamics is an analogy, not a derivation.

# Chapter 9

# Computational Universality and Subsumption

## 9.1 Turing Machine Embedding

**Theorem 9.1** (Abstract Simulation (Embedding_TM.v — reference-only, modular_proofs/TM_to_Minsky.v — reference-only)). *For every Turing Machine $T$, there exists a Minsky counter machine $M$ and a simulation relation $R$ such that $M$ simulates $T$ step-by-step, encoding the tape into two integers:*

$$\text{Left} = \sum_{i=0}^{h-1} t[h-1-i]\, 2^i, \qquad \text{Right} = \sum_{i=0}^{n-h-1} t[h+i]\, 2^i$$

*The Thiele Machine simulates Minsky using its reversible ALU. Hence:*

$$\text{TM} \preceq \text{Minsky} \preceq \text{Thiele}$$

## 9.2 Strict Containment

**Theorem 9.2** (Turing $\subsetneq$ Thiele (kernel/Subsumption.v)). *Classical Turing computation is **properly extended** by sighted Thiele computation (both are Turing-complete, but Thiele carries additional partition and $\mu$-accounting structure):*

$$\exists\, p.\ \text{sighted}(p) \wedge \neg\text{turing}(p)$$

*There exist programs that are Thiele-computable (using partition structure and $\mu$-accounting) but are not classical Turing programs. The extra structure is genuine new content.*

**Theorem 9.3** (Semantic Strictness (ThieleFoundations.v — reference-only)). *There exist Thiele traces with identical Turing shadows but non-isomorphic Thiele-level behavior. The partition and $\mu$ layers carry information not present in the Turing skeleton.*

## 9.3 Halting Undecidability

**Theorem 9.4** (Diagonal Argument (kernel/OracleImpossibility.v — reference-only)). *No total computable function decides the halting problem for the Thiele Machine.*

## 9.4 Oracle Cost Lower Bound

**Theorem 9.5** (Oracle Impossibility (kernel/OracleImpossibility.v — reference-only)). *The formalization defines a* sound *oracle as one charging $\Delta\mu \geq 1$ per query (`sound_oracle_cost`). Given this definition:*

1. *`oracle_halts_costs_mu`: A zero-cost oracle violates the soundness definition ($0 \geq 1 \Rightarrow \bot$).*

2. *`oracle_cost_linear`: Given `sound_oracle_cost`, $n$ queries cost $\geq n$.*

*These results verify the* internal consistency *of the pricing definition. The physical justification for why oracle queries should cost $\mu > 0$ comes from the No Free Insight framework (gaining information about undecidable propositions should cost something), but the Coq proofs verify the narrower claim that the pricing is self-consistent.*

## 9.5 Confluence

**Theorem 9.6** (Church–Rosser (Confluence.v — reference-only))**.** *For any state $s$ and two independent certificates $c_1, c_2$ (targeting distinct CNF formulas), applying them in either order yields the same $\mu$-cost:*

$$\mu(s_{12}) = \mu(s_{21})$$

*Independent structural operations commute.*

## 9.6 Tensoriality

**Theorem 9.7** (Module Independence (ThieleUnificationTensor.v — reference-only))**.** *For distinct modules $m_1 \neq m_2$, recording discoveries commutes:*

$$\mathrm{record}(\mathrm{record}(G, m_1, e_1), m_2, e_2) \equiv_{\mathrm{ext}} \mathrm{record}(\mathrm{record}(G, m_2, e_2), m_1, e_1)$$

*Local per-module operations are tensorial.*

## 9.7 Amortized Analysis

**Theorem 9.8** (Amortization (AmortizedAnalysis.v — reference-only))**.** *For a batch of $T$ instances with discovery cost $D$ and operational cost $O$ per instance:*

$$\text{total\_cost} = B \cdot D + T \cdot O \geq T \cdot O$$

*where $B$ is batch count. As $T \to \infty$, average cost $\to O$ (discovery overhead amortizes away).*

## 9.8 $\mu$-Complexity Classes

**Definition 9.1** ($\mu$P and $\mu$NP (kernel/MuComplexity.v))**.** Let PolyBound be a polynomial $p : \mathbb{N} \to \mathbb{N}$.

- A decision problem $f$ is in $\mu$**P** iff there exists a polynomial $p$ and a trace family such that for every input state of size $n$, the trace solves $f$ with $\mu$-cost $\leq p(n)$ and runs in polynomial time.

- A decision problem $f$ is in $\mu$**NP** iff there exists a polynomial $p$ and a verifier trace family such that every YES instance of size $n$ has a certificate verifiable with $\mu$-cost $\leq p(n)$ in polynomial time.

**Theorem 9.9** (Classical $P \subseteq \mu P$ (kernel/MuComplexity.v) — **(S)**)**.** *`classical_in_muP`: Any classically polynomial-time decision problem (using zero-cost instructions) is in $\mu P$ with $\mu$-cost $= 0$ for all inputs. Classical computation is a degenerate case of $\mu$-computation with no structural cost. Zero Admitted.*

**Remark 9.1.** $\mu$P and $\mu$NP separate the time cost of computation from the structural cost of certification. Whether $\mu\mathrm{P} \neq \mu\mathrm{NP}$ — or whether there exist problems in $\mu$NP not in $\mu$P — is an open question. The factored-search witness in `StructuralAdvantage.v` gives a concrete example of a $\mu$-cost advantage (paying $\mu = 18$ to certify a decomposition and reduce search from $O(N^2)$ to $O(2N)$), but that is not a class separation.

**Theorem 9.10** ($\mu$-Cost Hierarchy (kernel/MuHierarchyTheorem.v) — **(S)**)**.** *For every $k \geq 1$, the $\mu$-dimension admits a strict hierarchy:*

1. ***Achievability**: there exists a trace from `init_state` with $\mu$-cost exactly $k$ that achieves `vm_certified = true` and `vm_mu` $= k$.*
2. ***Lower bound**: any trace from `init_state` achieving `vm_mu` $\geq k$ must have spent at least $k$ units of $\mu$-cost.*

*Formally (`mu_hierarchy_theorem`): for every $k \geq 1$, the witness trace* $[$`instr_certify`$(k-1)]$ *(cost $= S(k-1) = k$) certifies `init_state` at level $k$, while no trace of $\mu$-cost $< k$ can reach `vm_mu` $\geq k$ from `init_state` (since `trace_mu_cost` $=$ `vm_mu_final` $- 0 \geq k$).*

***Corollary** (`mu_hierarchy_no_upper_bound`): no fixed $\mu$-budget suffices for all levels. The $\mu$-dimension is unbounded and non-collapsible. Zero Admitted.*

**Remark 9.2** ($\mu$-Hierarchy as the formal P-vs-NP analogue)**.** Theorem 9.10 is the $\mu$-analogue of the classical time hierarchy theorem. It formalizes the intuition behind $\mathrm{P} \neq \mathrm{NP}$ in the certified-structure setting: *finding* a certificate of structural complexity $k$ requires paying $\mu \geq k$, while *verifying* it (given the trace) costs 0 additional $\mu$. The hierarchy is strict, infinite, and proven from first principles with zero `Admitted` — no oracle, no diagonalization, no new axioms. The proof uses only `vm_apply_mu` (cost $= \mu$ increase) and `init_state_mu_zero` ($\mu$ starts at 0).

# Chapter 10

# $\mu$-Conservation and Receipt Integrity

## 10.1 The Conservation Law

**Theorem 10.1** ($\mu$-Ledger Conservation (MuLedgerConservation.v)). *For any trace* $\tau = [i_0, \ldots, i_k]$*:*

$$\mu_{\text{final}} = \mu_{\text{init}} + \sum_{j=0}^{k} \text{cost}(i_j)$$

*Every instruction increases* $\mu$ *by exactly its declared cost. No axioms, no admits.*

**Corollary 10.2** (Monotonicity). $\mu$ *never decreases:* $\mu(s) \leq \mu(\text{Run}(\tau, s))$ *for all* $\tau$*.*

**Corollary 10.3** (Irreversibility Bound). $\text{irreversible\_bits}(\tau) \leq \mu_{\text{final}} - \mu_{\text{init}}$.

**Theorem 10.4** ($\mu$ Decomposition (MuLedgerConservation.v)). $\mu_{\text{total}} = \mu_{\text{blind}} + \mu_{\text{sighted}}$ *for any partition of the cost into blind (reversible) and sighted (structural) components.*

## 10.2 Receipt Integrity

**Definition 10.1** (Receipt). A receipt $r = (\text{step}, \text{instr}, \mu_{\text{pre}}, \mu_{\text{post}}, h_{\text{pre}}, h_{\text{post}})$ records one transition.

**Theorem 10.5** (Receipt Integrity (ReceiptIntegrity.v)). *A valid receipt or execution attestation proves* $\mu_{\text{final}} = \mu_{\text{init}} + \sum \text{costs}$ *for the attested run. Any receipt claiming* $\Delta\mu \neq \text{cost}(\text{instr})$ *fails validation.*

**Theorem 10.6** (Non-Forgeability (ReceiptIntegrity.v)). *Forged receipts (claiming incorrect* $\Delta\mu$*) fail validation. Overflow values (*$\mu > 2^{31} - 1$*) are rejected at the receipt layer.*

# Chapter 11

# Gauge Symmetry and Noether's Theorem

## 11.1 The $\mathbb{Z}$-Action (Gauge Group)

**Definition 11.1** (Gauge Shift)**.** For $\delta \in \mathbb{Z}$, the gauge shift $\sigma_\delta : \mathcal{S} \to \mathcal{S}$ maps $S \mapsto S[\mu \leftarrow \mu + \delta]$, preserving all other fields.

**Theorem 11.1** (Group Action (KernelNoether.v))**.** *The gauge shifts form a $\mathbb{Z}$-action on states:*

1. ***Identity**: $\sigma_0(S) = S$.*
2. ***Composition**: $\sigma_a(\sigma_b(S)) = \sigma_{a+b}(S)$.*
3. ***Inverse**: $\sigma_{-n}(\sigma_n(S)) = S$ (when $\mu + n \geq 0$).*

## 11.2 Gauge Invariance

**Definition 11.2** (Observable)**.** Obs($S$) extracts the partition region structure, ignoring $\mu$.

**Theorem 11.2** (Gauge Invariance (KernelNoether.v))**.** $\mathrm{Obs}(\sigma_\delta(S)) = \mathrm{Obs}(S)$. *Shifting $\mu$ by a constant does not affect observables.*

**Theorem 11.3** (Noether's Theorem (KernelNoether.v, RepresentationTheorem.v — reference-only))**.**

1. ***Forward**: States identical except for $\mu$ lie on the same gauge orbit. (This is direct from the definition: if all other fields match, the $\mu$ difference is the gauge shift.)*
2. ***Gauge Invariance**: `z_gauge_invariance` proves that shifting $\mu$ does not change `Observable_partition`. This is definitional: `z_gauge_shift` only modifies the `vm_mu` field, and `Observable_partition` reads only `vm_graph`.*
3. ***Trace Preservation**: Gauge-equivalent states produce identical observable traces (same labels, same $\mu$-costs) for any finite horizon. Proven as `vm_step_orbit_equiv`.*

*The $\mu$-ledger is a gauge degree of freedom. Its* absolute *value is unobservable; only* differences *$\Delta\mu$ are physical. The "Noether" label is a deliberate analogy: the result follows from the record structure (separate fields for $\mu$ and graph), not from a deep symmetry principle.*

**Theorem 11.4** ($\mu$-Monotonicity (KernelNoether.v))**.** $\mathrm{vm\_step}(S, i, S') \implies \mu(S) \leq \mu(S')$. *The $\mu$-ledger never decreases under the dynamics. This is true by construction: instruction costs are natural numbers ($\geq 0$), so adding them to the ledger cannot decrease it. The proof examines each step constructor and confirms the arithmetic.*

# Chapter 12

# Two-Dimensional Amplitude Space

**Remark 12.1** (Epistemological Status: **(C)** Conditional Derivation)**.** The following derivation assumes that partition states admit *amplitude* representations (superpositions), not merely classical probability distributions. This is stated as Axiom 12.1. Without it, classical probability theory ($p \in [0,1]$, one-dimensional, continuous) suffices and the 2D argument does not apply. The passage from a discrete binary partition to the continuum $S^1$ also requires a limiting process not formalized in the Coq suite. The argument does not rule out quaternionic (4D) amplitudes; it establishes 2D as the *minimum* dimension for continuous superposition.

**Axiom 12.1** (Superposition Principle)**.** Partition module states admit amplitude representations: a state with $n$ classical configurations is described by $n$ real amplitudes $(a_1, \ldots, a_n)$ satisfying $\sum_i a_i^2 = 1$, where $a_i^2$ gives the probability of configuration $i$.

Given this axiom, binary partition structure forces quantum amplitudes to live in at least two dimensions.

**Theorem 12.1** (1D Is Insufficient (TwoDimensionalNecessity.v — reference-only))**.** *A one-dimensional normalized amplitude satisfies $x^2 = 1$, hence $x \in \{+1, -1\}$. No intermediate superpositions exist.*

**Theorem 12.2** (2D Is Necessary and Sufficient (TwoDimensionalNecessity.v — reference-only))**.** *Binary partition structure requires exactly 2D amplitude space. Two-dimensional amplitudes $(a, b)$ with $a^2 + b^2 = 1$ admit a continuous family via $a = \cos\theta$, $b = \sin\theta$: the unit circle $S^1$.*

# Chapter 13

# Complex Amplitudes from Norm Preservation

Zero-cost evolution must preserve the norm $a^2 + b^2 = 1$. The group of norm-preserving maps on $S^1$ is $\mathrm{SO}(2) \cong U(1)$, forcing amplitudes to be complex numbers.

**Theorem 13.1** (Rotation Group (ComplexNecessity.v — reference-only))**.** *2D rotations* $R_\theta(a, b) = (a\cos\theta - b\sin\theta,\ a\sin\theta + b\cos\theta)$ *satisfy:*

1. $|R_\theta(a, b)|^2 = a^2 + b^2$ *(norm preservation).*
2. $R_0 = \mathrm{id}$ *(identity).*
3. $R_{\theta_1} \circ R_{\theta_2} = R_{\theta_1+\theta_2}$ *(composition = addition).*
4. $R_\theta^{-1} = R_{-\theta}$ *(invertibility).*

**Theorem 13.2** (Complex Necessity (ComplexNecessity.v — reference-only))**.** *Complex multiplication by* $e^{i\theta}$ *is exactly 2D rotation. The norm-preserving maps on* $S^1$ *are exactly complex multiplications:* $\mathrm{SO}(2) \cong U(1)$*. Hence complex amplitudes are the* minimal *structure consistent with continuous 2D superposition; higher-dimensional extensions (quaternionic, etc.) are not ruled out by this argument alone.*

**Corollary 13.3** (Euler's Formula)**.** $e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$.

**Derivation chain**: Binary partition $\to$ 2D amplitudes $\to$ normalization ($S^1$) $\to$ zero-cost $\Rightarrow$ norm-preserving $\Rightarrow$ $\mathrm{SO}(2) \cong U(1)$ $\Rightarrow$ complex numbers.

# Chapter 14

# The Born Rule

**Definition 14.1** (Bloch Sphere Probabilities)**.** For a state with purity vector $(x, y, z)$:

$$P(|0\rangle) = \frac{1+z}{2}, \qquad P(|1\rangle) = \frac{1-z}{2}$$

**Definition 14.2** (Measurement $\mu$-Cost)**.**

$$\text{Cost}(x, y, z) = \frac{1 - (x^2 + y^2 + z^2)}{2}$$

This is the linear entropy. For pure states ($r^2 = 1$), cost $= 0$. For mixed states ($r^2 < 1$), cost $> 0$.

**Theorem 14.1** (Uniqueness of the Born Rule (BornRule.v))**.** *The Born rule $P(|0\rangle) = (1 + z)/2$ is the* ***unique*** *probability assignment satisfying:*

1. *Non-negativity and normalization: $P \geq 0$, $P(|0\rangle) + P(|1\rangle) = 1$.*
2. *Eigenstate consistency: $P(0, 0, 1) = 1$, $P(0, 0, -1) = 0$.*
3. *Linearity in $z$: $P(x, y, z) = az + b$ for some $a, b$.*

*Proof: Boundary conditions force $a = 1/2$, $b = 1/2$. No other solution exists.*

***Note:*** *The Coq theorem also lists a `mu_consistent` hypothesis, but it is* ***unused*** *in the proof. The hypothesis `mu_consistent P` unfolds to "measurement cost $\geq 0$ for valid Bloch vectors," which is directly from the definition true for all states and does not constrain $P$. The linearity assumption (item 3) does all the work—it assumes the functional form $P = az + b$ and solves a 2-equation system. This is less "deriving the Born rule from $\mu$-accounting" and more "the Born rule is the unique linear rule consistent with eigenstates." The non-circular derivation from tensor consistency (`BornRuleFromSymmetry.v`, below) is the stronger result.*

**Theorem 14.2** (Born Rule from Tensor Consistency (BornRuleFromSymmetry.v — reference-only/replaced; active derivation in `kernel/BornRuleLinearity.v`))**.** *Let $g : [0, 1] \to [0, 1]$ be any function satisfying:*

1. *Eigenstate consistency: $g(0) = 0$, $g(1) = 1$.*
2. *Normalization: $g(x) + g(1 - x) = 1$ for all $x \in [0, 1]$.*
3. *Tensor product consistency: $g(xy) = g(x) \cdot g(y)$ for $x, y \in [0, 1]$.*
4. *Range: $0 \leq g(x) \leq 1$ for $x \in [0, 1]$.*

*Then $g(x) = x$ for all $x \in [0, 1]$.*

**Remark 14.1** (Epistemological Status: **(C)** Conditional Derivation — Circularity Resolved)**.** The original proof (`BornRule.v`) depends on the linearity axiom (item 3): $P$ is affine in $z$. This assumption is equivalent to the Born rule restated, raising a circularity concern.

The new derivation (`BornRuleFromSymmetry.v`) replaces linearity with *tensor product consistency*: independent measurements yield independent outcomes, i.e., $g(xy) = g(x)g(y)$. This is a structural property of the machine, not a physics axiom. It follows from module independence proven in `ThieleUnificationTensor.v`. The proof proceeds: $g = \mathrm{id}$ on all dyadic rationals (by multiplicativity + normalization); then monotonicity + density of dyadics forces $g = \mathrm{id}$ everywhere (via the Archimedean property of $\mathbb{R}$). Zero admits.

# Chapter 15

# Unitarity

**Definition 15.1** (Evolution)**.** An evolution $E$ maps Bloch vectors $(x, y, z) \mapsto (x', y', z')$ with associated $\mu$-cost.

**Theorem 15.1** (Unitarity from Zero Cost (Unitarity.v))**.**

1. *$\mu = 0 \implies r'^2 \geq r^2$ (purity cannot decrease at zero cost). Proven as* `zero_cost_preserves_purity`*.*
2. *Non-unitary evolution ($r'^2 < r^2$ for some state) requires $\mu > 0$. Proven as* `nonunitary_requires_mu`*: the proof specializes the conservation hypothesis at the witness, then* `lra`*.*
3. *$\mu = 0 \Rightarrow$ no info loss (one direction proven). The converse and the full equivalence $\mu = 0 \Leftrightarrow$ unitary $\Leftrightarrow$ reversible are* stated *in the file but not proven as theorems: they appear as* `Definition` *declarations (propositional constants) rather than proven* `Theorem`*s.*

**Theorem 15.2** (Lindblad Requires $\mu$ (Unitarity.v))**.** *Lindblad-type dissipation at rate $\gamma > 0$ requires $\mu \geq \gamma$. Decoherence is paid for.*

**Theorem 15.3** (CPTP Structure (Unitarity.v))**.** *Positivity + trace preservation $\implies$ the evolution is CPTP (completely positive, trace-preserving).*

# Chapter 16

# Purification

**Theorem 16.1** (Purification Principle (Purification.v))**.** *Every mixed state* $(x, y, z)$ *with* $r^2 = x^2 + y^2 + z^2 \leq 1$ *admits eigenvalues* $\lambda_1, \lambda_2$ *with:*

- $\lambda_1 + \lambda_2 = 1, \quad 0 \leq \lambda_i \leq 1.$
- $(\lambda_1 - \lambda_2)^2 = r^2.$
- *Construction:* $\lambda_1 = (1 + r)/2$, $\lambda_2 = (1 - r)/2$.

*Pure states* ($r = 1$) *need no external reference (deficit* $= 0$*). The maximally mixed state* ($r = 0$) *has deficit* $= 1$.

# Chapter 17

# No-Cloning

**Theorem 17.1** (No-Cloning from $\mu$-Conservation (NoCloning.v)). *Let $I = x^2 + y^2 + z^2$ be the information content (purity). For a cloning operation with outputs of information $I_1, I_2$ and $\mu$-cost $\mu$:*

$$I_1 + I_2 \leq I + \mu$$

**Theorem 17.2** (No-Cloning from $\mu$-Monotonicity (NoCloningFromMuMonotonicity.v — reference-only/replaced; active proof in `kernel/NoCloning.v`)). *Machine-native no-cloning using nat-valued structural content (MDL complexity). If cloning duplicates $n$ bits of structural content while producing zero new bits: $2n \leq n + 0$ implies $n \leq 0$. The entire proof is a single `lia` step (linear integer arithmetic). No Bloch sphere, no Hilbert space, no real analysis. Zero admits.*

**Corollary 17.3** (Perfect Cloning Is Impossible at Zero Cost). *$I_1 = I_2 = I$ and $\mu = 0$ implies $2I \leq I$, hence $I \leq 0$. Only the trivial state can be cloned for free.*

**Corollary 17.4** (Cloning Cost). *Perfect cloning requires $\mu \geq I$. For pure states: $\mu \geq 1$.*

**Theorem 17.5** (Approximate Cloning (NoCloning.v)). *For fidelities $f_1, f_2$: $f_1 + f_2 \leq 1 + \mu/I$. At zero cost: $f_1 + f_2 \leq 1$.*

**Theorem 17.6** (No Deletion (NoCloning.v)). *Perfect deletion also requires $\mu \geq I$ (dual of no-cloning).*

# Chapter 18

# Observation and Collapse

## 18.1 Observation Irreversibility

**Theorem 18.1** (REVEAL Is Irreversible (ObservationIrreversibility.v — reference-only))**.** *For* bits $> 0$*:* $\mu_{\text{after}} > \mu_{\text{before}}$*, and the post-REVEAL state* $\neq$ *the pre-REVEAL state. Observation prevents recovery of the pre-measurement superposition.*

## 18.2 Collapse Determination

**Definition 18.1** (Partition Entropy)**.** $H(P) = \log_2(\dim(P))$ where dim is the partition state dimension.

**Theorem 18.2** (Maximum Information Implies Unique State (CollapseDetermination.v — reference-only))**.** *If bits revealed* $= H(P_{\text{before}})$ *(maximum information), then* $\dim(P_{\text{after}}) = 1$*. The measurement fully collapses the state to a unique outcome. This is the* ***projection postulate****, derived rather than assumed.*

# Chapter 19

# The Tsirelson Bound

**Definition 19.1** (Correlation $\mu$-Cost)**.** $\mu_{\text{corr}}(S) = 0$ if $|S| \leq 2\sqrt{2}$; otherwise $\mu_{\text{corr}} = |S| - 2\sqrt{2}$.

**Remark 19.1** (Epistemological Status: **(C)** Conditional Derivation — Circularity Resolved)**.** The original Coq proof (`TsirelsonDerivation.v` — reference-only/replaced) encoded $2\sqrt{2}$ in the definition of $\mu_{\text{corr}}$, making its derivation status **(R)**.

This gap is now closed. `TsirelsonGeneral.v` derives $S^2 \leq 8$ from pure algebra: a sum-of-squares identity proves each CHSH row contributes $\leq 2(e_1^2 + e_2^2)$, then Cauchy–Schwarz bounds the sum by $4\sum e_i^2 \leq 8$. `TsirelsonFromAlgebra.v` provides a self-contained algebraic derivation with achievability witness ($e = \pm 1/\sqrt{2}$, giving $S = 2\sqrt{2}$) and a verified rational bound ($\sqrt{8} < 5657/2000$). Zero admits in both files.

**Theorem 19.1** (Tsirelson from Zero $\mu$ (TsirelsonDerivation.v — reference-only/replaced) — **(R)**)**.** *Total $\mu = 0$ implies $|S_{\text{CHSH}}| \leq 2\sqrt{2} \approx 2.828$.*

**Theorem 19.2** (CHSH Separation (Deliverable_CHSHSeparation.v — reference-only))**.** *Strict numerical separation:*

$$2 \quad < \quad 2\sqrt{2} \approx 2.828 \quad < \quad \frac{16}{5} = 3.2$$

- ***Classical*** *(local receipts): $|S| \leq 2$ (Bell bound).*
- ***Quantum*** *(admissible, $\mu = 0$): $|S| \leq 2\sqrt{2}$ (Tsirelson bound).*
- ***Supra-quantum*** *($\mu > 0$ allowed): witnesses achieve $|S| = 3.2$.*

*The Tsirelson bound $2\sqrt{2}$ is the maximum correlation purchasable at zero $\mu$-cost. Exceeding it requires paying for additional structural information.*

# Chapter 20

# Witness Insight Taxonomy

**Theorem 20.1** (CHSH Trial Is Tier 0 (kernel/WitnessInsightGeneral.v) — **(S)**). *`chsh_trial_is_tier0`: The CHSH_TRIAL instruction never touches either certification channel:*

- *`chsh_trial_not_cert_addr_setter`: CHSH_TRIAL does not change `csr_cert_addr`.*
- *`chsh_trial_preserves_vm_certified`: CHSH_TRIAL does not change `vm_certified`.*

*Recording CHSH trial outcomes is provably free of structural cost. The counters go up; the cert channels are untouched. Cost: 0. Zero Admitted.*

**Theorem 20.2** (Three-Tier Observation Taxonomy (kernel/WitnessInsightGeneral.v) — **(S)**). *`witness_insight_complete_taxonomy`: Every observation the machine can make falls into exactly one of three tiers:*

**Tier 0** *Raw observation (always free): Records data without activating any certification channel. CHSH_TRIAL is the canonical example. Cost: $\Delta\mu = 0$ is possible; no cert channel is touched.*

**Tier 1** *Certified structural insight (always costs $\geq 1$): Any instruction that activates `csr_cert_addr` or `vm_certified` is in this tier. Proven: `certified_witness_insight_nonfree` shows $\mu$ increases by at least 1.*

**Tier 2** *Certified non-local witness insight (always costs $\geq 1$): A certified execution whose WitnessCount statistics show $S > 2$. `nonlocal_witness_insight_nonfree`: getting certified with non-local evidence requires $\mu \geq 1$.*

*`witness_insight_nonfree_general`: For any instruction in Tier 1 or Tier 2, $\mu$ increases by at least 1. The only free observations are Tier 0. Zero Admitted.*

# Chapter 21

# Bell's Theorem in Morphism Coupling Language

**Definition 21.1** (Setting-Outcome Coupling (kernel/CHSHCouplingBridge.v)). The `chsh_setting_outcome_c` maps a WitnessCount record to a coupling (list of $(nat \times nat)$ pairs). Each pair records a (setting-pair-index, outcome-type) observation. A coupling is **separable** (functional) when it maps each setting to at most one outcome type; **non-separable** otherwise.

**Theorem 21.1** (Locally Consistent Strategy Gives Separable Coupling (kernel/CHSHCouplingBridge.v) — **(S)**). *`locally_consistent_gives_separable_coupling`: Any WitnessCount configuration produced by a locally deterministic strategy $(a_0, a_1, b_0, b_1) \in \{0, 1\}^4$ gives a separable (functional) coupling. Zero Admitted.*

**Theorem 21.2** (Local Coupling Bound (kernel/CHSHCouplingBridge.v) — **(S)**). *`locally_consistent_classi` Any locally factorizable morphism coupling satisfying `WCLocallyConsistent` has $|S| \leq 2$. Bell's theorem in the morphism coupling language: the separation between local and non-local correlations is not just a statistical fact but a structural property of the coupling predicate. Zero Admitted.*

**Theorem 21.3** (CHSH Violation Rules Out Local Couplings (kernel/CHSHCouplingBridge.v) — **(S)**). *`chsh_violation_rules_out_locally_factorizable_coupling`: If WitnessCount statistics satisfy $S > 2$, no locally factorizable morphism coupling is consistent with those counts. Contrapositively: non-local correlations cannot be produced by any local morphism coupling. The no-go is stated in the coupling language, not just the statistical layer. Zero Admitted.*

**Remark 21.1.** These results connect the categorical morphism layer directly to Bell nonlocality. A claim that two modules are entangled (via MORPH_ASSERT on a non-separable coupling) cannot be locally faked: the NoFI law charges at least 1 for the MORPH_ASSERT, and the coupling structure itself rules out any local hidden strategy that would produce $S > 2$ for free.

# Chapter 22

# The Schrödinger Equation

**Theorem 22.1** (Emergent Schrödinger (EmergentSchrodinger.v — reference-only))**.** *The finite-difference Schrödinger equation emerges from the partition update rules. For a two-component amplitude state* $(a, b)$ *with mass* $m$*, potential* $V$*, and time step* $\Delta t$*:*

$$a(t + \Delta t) = a - \frac{\Delta t}{2m}\nabla^2 b + \Delta t \cdot Vb \tag{22.1}$$

$$b(t + \Delta t) = b + \frac{\Delta t}{2m}\nabla^2 a - \Delta t \cdot Va \tag{22.2}$$

*The coupling is* ***antisymmetric****:* $c_{a,\nabla^2 b} = -c_{b,\nabla^2 a}$ *and* $c_{a,Vb} = -c_{b,Va}$*, which is necessary and sufficient for probability conservation* $\partial_t(a^2 + b^2) = 0$*.*

**Remark 22.1** (Epistemological Status: **(C)** Conditional Derivation)**.** This is exactly the finite-difference discretization of $i\hbar\partial_t\psi = -\frac{\hbar^2}{2m}\nabla^2\psi + V\psi$ with $\psi = a + ib$. The Schrödinger form emerges *conditionally*: given Axiom 12.1 (which entails 2D amplitudes) and the requirement that zero-cost evolution preserves probability ($\partial_t(a^2 + b^2) = 0$), the antisymmetric coupling structure is the unique linear solution. The mass $m$ and potential $V$ are parameters, not derived.

# Chapter 23

# Planck's Constant

**Remark 23.1** (Epistemological Status: **(R)** Consistency Relation)**.** The following is **not** a derivation of Planck's constant from first principles. It is a consistency relation that gives $h$ a physical interpretation within the $\mu$-framework. The Coq proof verifies an algebraic identity; the physics is in the definitions.

**Definition 23.1** (Physical Constants (Normalized Units))**.** $k_B = 1/100, \quad T = 1, \quad h = 1$. Landauer energy: $E_L := k_B T \ln 2$.

**Definition 23.2** (Computational Time Step)**.** $\tau_\mu := h/(4E_L)$, the Margolus–Levitin time at Landauer energy.

**Theorem 23.1** (Planck Consistency Relation (PlanckDerivation.v — reference-only) — **(R)**)**.**

$$h = 4 \cdot E_L \cdot \tau_\mu$$

**Remark 23.2** (Why This Is Circular)**.** Substituting $\tau_\mu = h/(4E_L)$ yields $h = 4E_L \cdot h/(4E_L) = h$. The Coq proof reduces to the `field` tactic (pure algebra). This does *not* predict $h$; it defines $\tau_\mu$ in terms of $h$ and verifies consistency.

**What the relation does provide**: a physical interpretation of Planck's constant as $4\times$ the action (energy $\times$ time) of one $\mu$-bit operation at Landauer cost. If taken as a physical claim, the testable content is the predicted time step $\tau_\mu = h/(4k_B T \ln 2)$, which at room temperature yields $\tau_\mu \approx 5.7 \times 10^{-14}$ s (femtosecond range, consistent with molecular vibration timescales).

**What would make this non-circular**: defining $\tau_\mu$ as an independent primitive (measured experimentally) and then *deriving* $h = 4E_L\tau_\mu$ as a *prediction.* The current formalization does not do this.

# Chapter 24

# Emergent Spacetime

## 24.1 The $\mu$-Metric

**Definition 24.1** ($\mu$-Distance (MuGeometry.v))**.** The Coq formalization defines `mu_distance` as the absolute difference of $\mu$-totals:

$$d_\mu(S_1, S_2) = |\mu_{\text{total}}(S_2) - \mu_{\text{total}}(S_1)|$$

This is a one-dimensional projection, not a minimum over paths.

**Theorem 24.1** (Pseudometric Properties (MuGeometry.v))**.** *$d_\mu$ satisfies non-negativity, reflexivity ($d(S, S) = 0$), symmetry, and triangle inequality, all inherited from absolute value on $\mathbb{Z}$. Note: $d_\mu$ is a **pseudometric**, not a metric: $d(S_1, S_2) = 0$ does* not *imply $S_1 = S_2$ (many distinct states share the same $\mu$-total). The "geometry" is a 1D projection onto the $\mu$ counter.*

## 24.2 Causal Cones

**Definition 24.2** ($\mu$-Reachability Cone (SpacetimeEmergence.v))**.** $C^+(S) = \{S' \mid \exists\tau.\ S \xrightarrow{\tau} S' \wedge \text{cost}(\tau) \leq \text{Budget}\}$.

This is a structural analogy to causal cones: the set of reachable states given a finite $\mu$-budget plays a role analogous to a light cone. The analogy should not be over-read: there is no Lorentz invariance, no metric signature, and no boost transformations in the Coq formalization.

## 24.3 No-Signaling

**Theorem 24.2** (Observational Locality (SpacetimeEmergence.v))**.** *If module $M_B$ is not in the target set of any instruction in a trace, then $M_B$'s observable region is unchanged after execution (`exec_trace_no_signaling_outside_cone`).*

**Remark 24.1.** This is a **data isolation** property: operations targeting module $A$ in the partition graph do not affect the lookup of module $B$. The proof is structural (list operations on disjoint keys), non-trivial mainly for `psplit`/`pmerge` which restructure the graph. The label "no-signaling" is an analogy—there is no spatial metric or speed limit between modules.

## 24.4 Four-Dimensional Spacetime Geometry

The previous sections treated spacetime as a 1D projection via the $\mu$-metric. The next sections extend to full 4D geometry with discrete differential structure, building discrete differential geometry from computational primitives.

### 24.4.1 4D Simplicial Complex

**Definition 24.3** (4D Simplicial Complex (FourDSimplicialComplex.v))**.** A **4D simplicial complex** $\mathcal{K}$ is a discrete approximation to 4D spacetime consisting of:

- **Vertices** $V$: Computational modules (ModuleID)
- **Edges** $E \subseteq V \times V$: Connections between modules
- **Simplices**: 2-simplices (triangles), 3-simplices (tetrahedra), 4-simplices (4D cells)

The complex satisfies:

- **Consistency**: Every face of a simplex is also in $\mathcal{K}$
- **Well-formedness**: Dimension 4 with locally Euclidean structure

### 24.4.2 Metric Tensor from $\mu$-Costs

**Definition 24.4** (Computational Metric (MetricFromMuCosts.v, RiemannTensor4D.v))**.** The metric tensor at vertex $v$ with respect to vertices $w_1, w_2$ is:

$$g_{\mu\nu}(v; w_1, w_2) = \begin{cases} \ell(v, w_1, w_2) & \text{if } \mu = \nu \\ 0 & \text{if } \mu \neq \nu \end{cases} \tag{24.1}$$

where the edge length function is:

$$\ell(s, v_1, v_2) = 2 \cdot \text{mass}(v_1) \quad \text{if } v_1 = v_2 \tag{24.2}$$

and mass is derived from structural complexity: $\text{mass}(v) = |\text{module\_structural\_mass}(s, v)|$.

**Remark 24.2.** This metric is **position-dependent** for non-uniform mass distributions:

- **Uniform mass** $m$: $g_{\mu\mu}(v; w, w) = 2m$ (constant) $\implies$ flat spacetime
- **Non-uniform mass**: $g_{\mu\mu}(v; w, w)$ varies with $w \implies$ curved spacetime

In this model, mass distributions create position-dependent geometry: uniform mass gives flat spacetime, non-uniform mass gives curved spacetime.

### 24.4.3 Discrete Differential Geometry

**Definition 24.5** (Discrete Derivative (RiemannTensor4D.v))**.** For a scalar field $f : V \to \mathbb{R}$ and direction $\mu$:

$$\partial_\mu f(v) = \frac{1}{|N_\mu(v)|} \sum_{w \in N_\mu(v)} (f(w) - f(v)) \tag{24.3}$$

where $N_\mu(v)$ is the set of neighbors of $v$ in the $\mu$-direction.

**Definition 24.6** (Christoffel Symbols (RiemannTensor4D.v))**.** The connection coefficients (Christoffel symbols of the second kind) are:

$$\Gamma^\rho_{\mu\nu}(v) = \frac{1}{2} \sum_\sigma g^{\rho\sigma}(v) \Big( \partial_\mu g_{\nu\sigma}(v) + \partial_\nu g_{\mu\sigma}(v) - \partial_\sigma g_{\mu\nu}(v) \Big) \tag{24.4}$$

where $g^{\rho\sigma}$ is the inverse metric (diagonal in this construction).

**Lemma 24.3** (Flat Spacetime Has Zero Connection (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass distribution:*

$$\forall w,\ mass(w) = m \quad \implies \quad \Gamma^\rho_{\mu\nu}(v) = 0 \tag{24.5}$$

*Proof.* Uniform mass $\implies$ position-independent metric $\implies \partial_\mu g_{\nu\sigma} = 0 \implies \Gamma^\rho_{\mu\nu} = 0$. □

### 24.4.4 Curvature Tensors

**Definition 24.7** (Riemann Curvature Tensor (RiemannTensor4D.v)). The Riemann curvature tensor measures the failure of parallel transport:

$$\begin{aligned} R^{\rho}_{\sigma\mu\nu}(v) &= \partial_\mu \Gamma^{\rho}_{\nu\sigma}(v) - \partial_\nu \Gamma^{\rho}_{\mu\sigma}(v) \\ &\quad + \sum_{\lambda} \Big( \Gamma^{\rho}_{\mu\lambda}(v)\Gamma^{\lambda}_{\nu\sigma}(v) - \Gamma^{\rho}_{\nu\lambda}(v)\Gamma^{\lambda}_{\mu\sigma}(v) \Big) \end{aligned} \tag{24.6}$$

**Definition 24.8** (Ricci Tensor (RiemannTensor4D.v)). Contraction of the Riemann tensor:

$$R_{\mu\nu}(v) = \sum_{\rho} R^{\rho}_{\mu\rho\nu}(v) \tag{24.7}$$

**Definition 24.9** (Ricci Scalar (RiemannTensor4D.v)). Complete contraction:

$$R(v) = \sum_{\mu,\nu} g^{\mu\nu}(v) R_{\mu\nu}(v) \tag{24.8}$$

**Definition 24.10** (Einstein Tensor (RiemannTensor4D.v)). The Einstein tensor encodes spacetime curvature:

$$G_{\mu\nu}(v) = R_{\mu\nu}(v) - \frac{1}{2} g_{\mu\nu}(v) R(v) \tag{24.9}$$

**Lemma 24.4** (Flat Spacetime Has Zero Curvature (EinsteinEquations4D.v) — **(S)**). *For uniform mass:*

$$\forall w,\ \mathit{mass}(w) = m \quad \Longrightarrow \quad R^{\rho}_{\sigma\mu\nu}(v) = 0 \quad \Longrightarrow \quad G_{\mu\nu}(v) = 0 \tag{24.10}$$

### 24.4.5 Stress-Energy Tensor

**Definition 24.11** (Energy Density (EinsteinEquations4D.v)). The energy density is the computational mass:

$$T_{00}(v) = \mathrm{mass}(v) \tag{24.11}$$

**Definition 24.12** (Momentum Density (EinsteinEquations4D.v)). Spatial derivatives of energy:

$$T_{0i}(v) = T_{i0}(v) = \partial_i T_{00}(v) \tag{24.12}$$

**Definition 24.13** (Stress Components (EinsteinEquations4D.v)). Spatial stress tensor:

$$T_{ij}(v) = \begin{cases} \mathrm{mass}(v) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{24.13}$$

**Definition 24.14** (Stress-Energy Tensor (EinsteinEquations4D.v)). Combining all components:

$$T_{\mu\nu}(s, \mathcal{K}, v) = \begin{cases} T_{00}(v) & \text{if } \mu = \nu = 0 \\ T_{0i}(v) & \text{if } \mu = 0, \nu = i \text{ or } \mu = i, \nu = 0 \\ T_{ij}(v) & \text{if } \mu = i, \nu = j \\ 0 & \text{otherwise} \end{cases} \tag{24.14}$$

### 24.4.6 Einstein's Field Equations

**Definition 24.15** (Gravitational Constant (EinsteinEquations4D.v))**.** In computational units where the fundamental scale is set by $\mu$-costs:

$$G = \frac{1}{8\pi} \tag{24.15}$$

This makes $8\pi G = 1$, normalizing the coupling in Einstein's equations.

**Lemma 24.5** (Curvature from $\mu$-Gradients (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass distribution:*

$$\forall w,\ \mathit{mass}(w) = m \quad \Longrightarrow \quad G_{\mu\nu}(v) = 0 \tag{24.16}$$

*Proof.* The proof chain is:

1. Uniform mass $\implies$ position-independent metric
2. Position-independent metric $\implies$ zero discrete derivatives $\partial_\mu g_{\nu\sigma} = 0$
3. Zero metric derivatives $\implies$ zero Christoffel symbols $\Gamma^\rho_{\mu\nu} = 0$
4. Zero Christoffel $\implies$ zero Riemann tensor $R^\rho_{\sigma\mu\nu} = 0$
5. Zero Riemann $\implies$ zero Ricci tensor and scalar
6. Zero Ricci $\implies$ zero Einstein tensor $G_{\mu\nu} = 0$

All steps are proven constructively in `EinsteinEquations4D.v`. □

**Lemma 24.6** (Vacuum Implies Zero Stress-Energy (EinsteinEquations4D.v) — **(S)**)**.** *For vacuum states (zero mass everywhere):*

$$\forall w,\ \mathit{mass}(w) = 0 \quad \Longrightarrow \quad T_{\mu\nu}(v) = 0 \tag{24.17}$$

*Proof.* Direct computation:

- $T_{00} = \text{mass} = 0$
- $T_{0i} = \partial_i(\text{mass}) = \partial_i(0) = 0$
- $T_{ij} = \delta_{ij} \cdot \text{mass} = \delta_{ij} \cdot 0 = 0$

□

**Theorem 24.7** (Einstein Equations in Vacuum (EinsteinEquations4D.v) — **(S)**)**.** *For any VM state $s$, 4D simplicial complex $\mathcal{K}$, spacetime indices $\mu, \nu$, and vertex $v$:*

$$\boxed{\forall w,\ \mathit{mass}(w) = 0 \quad \Longrightarrow \quad G_{\mu\nu}(s, \mathcal{K}, v) = 8\pi G\, T_{\mu\nu}(s, \mathcal{K}, v)} \tag{24.18}$$

***Note:*** *This is the vacuum case where both sides equal zero ($0 = 0$). The gravitational constant $G := 1/(8\pi)$ is a **unit choice** (a definition), not a derived quantity. In vacuum, all masses vanish, so $T_{\mu\nu} = 0$ and uniform-mass (zero) yields $G_{\mu\nu} = 0$; the equation reduces to $0 = 8\pi \cdot \frac{1}{8\pi} \cdot 0 = 0$. The non-vacuum case is **not proven** and would require discrete Bianchi identities not present in the formalization.*

*Proof.* **Step 1**: Expand the right-hand side using $G = 1/(8\pi)$:

$$8\pi G\, T_{\mu\nu} = 8\pi \cdot \frac{1}{8\pi} \cdot T_{\mu\nu} = T_{\mu\nu} \tag{24.19}$$

**Step 2**: Apply Lemma 24.5 (curvature from $\mu$-gradients):
Vacuum (mass = 0 everywhere) is a special case of uniform mass distribution, so:

$$G_{\mu\nu}(v) = 0 \tag{24.20}$$

**Step 3**: Apply Lemma 24.6 (stress-energy conservation):
For vacuum:

$$T_{\mu\nu}(v) = 0 \tag{24.21}$$

**Conclusion**: Both sides equal zero:

$$0 = 0 \quad \checkmark \tag{24.22}$$

The complete proof is formalized in `coq/kernel/EinsteinEquations4D.v` with:

- No project-local axioms
- Zero admitted lemmas
- Zero classical logic imports
- Verified by automated proof auditing (Inquisitor: 0 HIGH findings)

□

**Remark 24.3** (Vacuum vs. Non-Vacuum)**.** Theorem 24.7 requires the vacuum hypothesis (mass = 0 everywhere). For non-vacuum states with arbitrary mass distributions, the equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ only holds when the distribution satisfies discrete Bianchi identities (conservation laws). This is analogous to classical general relativity, where Einstein's equations are *field equations* that constrain which configurations are physically possible, not universal identities satisfied by all states.

The vacuum case is sufficient to demonstrate that spacetime curvature emerges from computational dynamics. The non-vacuum case requires discrete differential geometry beyond the current formalization (specifically, discrete covariant derivatives and discrete Bianchi identities).

**Remark 24.4** (Epistemological Status)**.** This is a **(S) Structural** result: it proves a mathematical relationship between computational primitives (VM states, $\mu$-costs) and geometric quantities (curvature tensors) with no physical axioms. The theorem establishes that:

1. Spacetime geometry (metric, curvature) can be constructed from pure computation
2. The resulting geometry satisfies the vacuum Einstein equation in the formally verified zero-mass regime
3. Beyond that vacuum regime, the repository contains a conditional full-tensor bridge with named premises, a proved uniform diagonal boundary witness in the active operator, and a separately proved non-uniform boundary witness in a concrete affine metric-scaled redesign branch rather than one universal unconditional theorem

The result does *not* claim that physical spacetime *is* implemented by a Turing machine—that would be a **(C)** conditional claim requiring empirical validation. More importantly, it does not justify the stronger statement that arbitrary computational substrates with $\mu$-cost tracking automatically satisfy the full Einstein equations in complete generality. The honest status is narrower: the vacuum theorem is unconditional; the full-tensor bridge in `EinsteinEquationsFull.v` remains conditional; the active first-neighbor operator proves the uniform diagonal boundary witness and formally refutes the stronger non-uniform diagonal extension; the broader non-negative local weighted family containing the simple symmetric average also fails on the same shell; and `SymmetricDerivative4D.v` proves that a concrete signed affine metric-scaled redesign rescues that non-uniform boundary witness with off-diagonal Ricci vanishing, Einstein tensor $G_{\mu\nu} = 3\,\delta_{\mu\nu}$ at the critical vertex, and, under `module_structural_mass s 1 = 1`, the aligned full-tensor witness $G_{\mu\nu} = 3\,T_{\mu\nu}$ there.

**Corollary 24.8** (Discrete Vacuum Geometry is Emergent)**.** *Discrete vacuum spacetime geometry is an emergent property of the machine's information-processing structure.*

*Proof.* Theorem 24.7 constructs the discrete metric, curvature, and vacuum Einstein relation from computational primitives (VM states, operations, $\mu$-costs) with no physical axioms. Therefore, this model's discrete vacuum geometry is a derived concept rather than a primitive one. □

### 24.4.7 Empirical Validation

The theoretical results are validated numerically:

**Theorem 24.9** (Numerical Verification (tests/test_*) — Empirical)**.**

1. ***Flat spacetime*** *(uniform mass $m = 1$):*
   - $\max|\Gamma^{\rho}_{\mu\nu}| < 10^{-10}$ ✓
   - $\max|R^{\rho}_{\sigma\mu\nu}| < 10^{-10}$ ✓
   - $\max|G_{\mu\nu}| < 10^{-10}$ ✓
2. ***Curved spacetime*** *(non-uniform mass $m \in [0.5, 1.5]$):*
   - $\max|\Gamma^{\rho}_{\mu\nu}| \approx 3.0$ ✓ *(curvature detected)*
   - $\max|R^{\rho}_{\sigma\mu\nu}| > 0$ ✓
3. ***Vacuum Einstein equations*** *(mass $= 0$):*
   - $|G_{\mu\nu}| < 10^{-10}$ ✓
   - $|T_{\mu\nu}| < 10^{-10}$ ✓
   - $|G_{\mu\nu} - 8\pi G T_{\mu\nu}| < 10^{-10}$ ✓

These tests confirm that the discrete differential geometry correctly implements the theoretical predictions.

### 24.4.8 Affine Metric-Scaled Operator: Non-Uniform Boundary Results

The first-neighbor discrete operator refutes the non-uniform diagonal extension of the EFE. A signed affine metric-scaled redesign closes the gap at vertex 1 and fully characterizes the outer vertices.

**Theorem 24.10** (Affine Off-Diagonal Ricci Zero (kernel/AffineEFEClosure.v) — **(S)**)**.** `affine_off_diagonal_r` *Under the affine metric-scaled symmetric operator, with the non-uniform isotropic boundary 4-simplex metric, off-diagonal Ricci vanishes at vertex 1: $R_{\mu\nu}(v_1) = 0$ for $\mu \neq \nu$. Zero Admitted. Zero Section Variables.*

**Theorem 24.11** (Full Tensor EFE at Vertex 1 (kernel/AffineEFEClosure.v) — **(S)**). `affine_full_tensor_efe`: *Under* `module_structural_mass s 1 = 1` *(unit structural mass at vertex 1), the full tensor Einstein field equation holds:*

$$G_{\mu\nu}(v_1) = 3 \cdot T_{\mu\nu}(v_1)$$

*for all* $\mu, \nu \in \{0, 1, 2, 3\}$. *Zero Admitted.*

**Theorem 24.12** (Outer Vertex Characterization (kernel/AffineEFEClosure.v) — **(S)**). *At the outer vertices* $v \in \{0, 2, 3, 4\}$ *of the boundary 4-simplex, the affine operator produces:*

- `affine_ricci_offdiag_outer`: *Off-diagonal Ricci* $= 3/32$ *(non-zero).*
- `affine_ricci_diag_outer`: *Diagonal Ricci* $= -45/32$.
- `affine_ricci_scalar_outer`: *Ricci scalar* $= -45/8$.
- `affine_einstein_outer`: *Einstein tensor* $= 45/32$ *(diagonal) and* $3/32$ *(off-diagonal).*
- `affine_efe_fails_outer_offdiag`: *The diagonal EFE form* $G = 3T$ *fails at outer vertices because* $G$ *has non-zero off-diagonal components.*

*The affine factor* $8 - 5g_{00}$ *is specifically tuned to the center vertex geometry; at outer vertices it produces a non-trivial off-diagonal Einstein tensor. Zero Admitted.*

**Remark 24.5** (What Remains Open). The affine operator closes vertex 1 and characterizes the outer vertices. The remaining open question is whether the affine factor $8 - 5g_{00}$ is the correct physical discretization. That requires connecting the discrete operator to a physical substrate — not a Coq question, an empirical one.

# Chapter 25

# Information Causality

**Theorem 25.1** (IC–$\mu$ Equivalence (InformationCausality.v) — **(R)**). *The file defines two record types (`ICScenario` and `MuScenario`) with opaque `Prop` fields, then proves they are equivalent* given *a hypothesis (`ic_mu_equivalent`) that literally states the equivalence. The proof is `tauto`. No mutual information, probability distributions, or CHSH values appear anywhere in the formalization.*

**Remark 25.1.** The intended statement — that Bob's total information about Alice's data is bounded by the channel's $\mu$-capacity — is a reasonable physical claim but is **not formalized** in the Coq file. The current file is a placeholder that assumes the conclusion as a hypothesis and re-derives it. A future version would need to define entropy, mutual information, and prove the bound from the VM semantics.

# Chapter 26

# Thermodynamic Bridge

## 26.1 Landauer's Principle

**Theorem 26.1** (Landauer Derived (LandauerDerived.v)).

1. `erasure_irreversible`: *Erasing* $\geq 1$ *bit implies fan-in* $= 2^k > 1$ *(arithmetic:* $2^k > 1$ *for* $k \geq 1$*).*

2. `erasure_decreases_entropy`: *Erasure decreases system entropy (*$\Delta S < 0$*) — this is just* $\text{output} < \text{input} \Rightarrow \text{output} - \text{input} < 0$.

3. `landauer_information_bound`: *For any physical erasure, environment entropy increase* $\geq$ *bits erased.*

4. `erasure_additive`: *Sequential erasures compose additively.*

***Caveat on item 3:*** *The* `PhysicalErasure` *record has* `second_law_satisfied` *as a* ***field****, not a derived property. The theorem* `landauer_information_bound` *simply extracts this field, making the bound an* input assumption *rather than a derivation. The physical energy bound* $Q \geq k_B T \ln 2 \cdot n$ *is* ***explicitly not proven*** *(the file itself notes the conversion factor is not formalized). Also, the* `mu` *in this file is a standalone synonym for* `bits_erased`*—it has no formal connection to* `vm_mu` *in* `VMState.v`*.*

## 26.2 Dissipative Embedding

**Theorem 26.2** (Dissipative Model (DissipativeEmbedding.v — reference-only))**.** *Irreversible (dissipative) physics embeds into the Thiele VM with* $\Delta\mu \geq 1$ *per irreversible step. Reversible physics embeds with* $\Delta\mu = 0$*.*

**Theorem 26.3** (Reversible Physics (PhysicsEmbedding.v — reference-only, WaveEmbedding.v — reference-only))**.** *Reversible lattice gas and wave models embed with* $\mu_{\text{final}} = \mu_{\text{init}}$ *(zero cost). Particle count, momentum, wave energy, and wave momentum are conserved by the VM.*

# Chapter 27

# Self-Reference and the Thiele Manifold

## 27.1 Self-Reference Escalation

**Theorem 27.1** (Gödel-Style Escalation (SelfReference.v)). *Any self-referential system $S$ requires a meta-system* Meta *with strictly more dimensions:*

$$\text{self\_ref}(S) \implies \exists\,\text{Meta} : \dim(\text{Meta}) > \dim(S)$$

*The meta-system inherits self-reference, creating an infinite escalation.*

## 27.2 The Thiele Manifold

**Definition 27.1** (Thiele Manifold (ThieleManifold.v)). An infinite tower of systems $\{L_n\}_{n\in\mathbb{N}}$ with:

- $\dim(L_n) = 4 + n$ (level $n$ has dimension $4 + n$).
- $\dim(L_{n+1}) > \dim(L_n)$ (strict enrichment).
- Each level reasons about the level below.
- Self-reference at level $n$ is answered by level $n + 1$.

**Theorem 27.2** (Projection to Spacetime (ThieleManifold.v)). *The projection $\pi_4$ collapses the tower to dimension 4 (spacetime). For $n > 0$:*

- *$\pi_4$ is lossy:* $\dim(L_n) > 4$.
- *$\mu$-cost of projection* $= \dim(L_n) - 4 = n > 0$.

*Spacetime is the "shadow" of the full manifold.*

## 27.3 Genesis: Process–Machine Isomorphism

**Theorem 27.3** (Genesis (Genesis.v — reference-only)). *There is a definitional isomorphism* Proc $\cong$ Thiele *between coherent processes (step + admissibility proof) and Thiele machines:*

$$\text{thiele\_to\_proc}(\text{proc\_to\_thiele}(P)) = P \tag{27.1}$$

$$\text{proc\_to\_thiele}(\text{thiele\_to\_proc}(T, H)) = T \tag{27.2}$$

*Any coherent process* is *a Thiele machine, and vice versa.*

# Chapter 28

# Three-Layer Isomorphism

**Theorem 28.1** (Observable Comparison Contract (ThreeLayerIsomorphism.v)). *The active Coq theorem packages the* observable-state contract *used for cross-layer comparison: if the chosen Python and RTL decoders satisfy the same projection contract as the Coq layer, then the decoded observable states agree on that projection for a shared trace $\tau$.*

$$\mathrm{decode}_{\mathrm{Python}}(\tau) = \mathrm{decode}_{\mathrm{Coq}}(\tau) \wedge \mathrm{decode}_{\mathrm{RTL}}(\tau) = \mathrm{decode}_{\mathrm{Coq}}(\tau) \Longrightarrow \mathrm{decode}_{\mathrm{Python}}(\tau) = \mathrm{decode}_{\mathrm{RTL}}(\tau)$$

*This is a contract theorem over explicit observable projections, not an unconditional claim that every current Python and RTL artifact is universally equal to Coq in all CI environments.*

## 28.1 RTL Coverage: Zero Gaps

**Theorem 28.2** (RTL Gap Registry Empty (kami_hw/RTLGapRegistry.v) — **(S)**). *rtl_gap_count: The RTL gap registry is empty. There are zero unresolved coverage gaps between the Kami hardware specification and the Coq kernel semantics:*

$$\mathit{rtl\_gap\_count} = 0$$

*rtl_coverage_partition: The 46-opcode ISA partitions as:*

$$36\ (\textit{unconditional Qed}) + 10\ (\textit{conditional Qed under inductive invariants}) + 0\ (\textit{gaps}) = 46$$

*This is a Coq theorem, not a comment. The partition equation is machine-checked. Zero Admitted.*

**Remark 28.1.** The 36 unconditional proofs cover opcodes through `SupportedOpcode`, plus CALL, RET, CHSH_TRIAL (`EmbedStep_WF.v`), TENSOR_SET, TENSOR_GET, LASSERT (`GraphReconstructionBridge.v`). The 10 conditional proofs cover the 4 partition opcodes and 7 morphism opcodes under `extended_hw_invariant / coupling_wf`, which are proved inductive: they hold at initialization and are preserved by every instruction in the ISA.

# Chapter 29

# The Curry–Howard–Thiele Correspondence

**Theorem 29.1** (Logic–Computation Isomorphism (LogicIsomorphism.v — reference-only))**.** *The Thiele Machine extends the Curry–Howard correspondence:*

| ***Logic*** | ***Computation*** | ***Thiele*** |
|---|---|---|
| *Proposition* | *Type* | *Partition* |
| *Proof* | *Program (term)* | *Trace* |
| *Cut elimination* | *$\beta$-reduction* | *Execution (*Run*)* |
| *Proof equivalence* | $=_\beta$ | *Execution equivalence* |

*A valid proof corresponds to a terminating execution. Proof equivalence $\Leftrightarrow$ execution equivalence.*

**Theorem 29.2** (Logic Is Physics (LogicToPhysics.v — reference-only))**.** *Cut elimination in logic = relational composition in physics:*

$$\text{interp}(\text{cut}(\pi_1, \pi_2)) = \text{rel\_comp}(\text{interp}(\pi_1), \text{interp}(\pi_2))$$

*Logical proof composition maps to physical relation composition. This is the formal nucleus of the claim that logic and physics share the same categorical structure.*

# Chapter 30

# Categorical Structure

## 30.1 The Morphism Layer

The active implementation adds a concrete category directly to the machine state. This is not an abstract analogy — it is an executable layer of 7 opcodes (0x27–0x2D) with formally verified laws.

### 30.1.1 Objects and Morphisms

The partition graph $G$ now carries two maps:

- `pg_modules` : ModuleID $\rightharpoonup$ ModuleState — objects (partition modules)
- `pg_morphisms` : MorphismID $\rightharpoonup$ MorphismState — arrows

A MorphismState bundles: source module, target module, coupling data (list of $(nat \times nat)$ pairs plus a label), and an identity flag.

### 30.1.2 The Seven Categorical Opcodes

- `MORPH`$(dst, src, tgt, idx, \Delta\mu)$: create morphism $src \to tgt$, write morphism ID to `regs`$[dst]$. Costs $\Delta\mu$.
- `COMPOSE`$(dst, f, g, \Delta\mu)$: compose morphisms $f; g$ when $f$.target $= g$.source, write new ID to `regs`$[dst]$. Costs $\Delta\mu$. Errors if endpoints mismatch.
- `MORPH_ID`$(dst, m, \Delta\mu)$: create identity morphism $\mathrm{id}_m$ for module $m$ (diagonal relation). Costs $\Delta\mu$.
- `MORPH_DELETE`$(f, \Delta\mu)$: remove morphism $f$. Cascades on module deletion via `graph_cascade_delete_mor`
- `MORPH_ASSERT`$(f, prop, cert, \Delta\mu)$: cert-setter. Costs $S(\Delta\mu) \geq 1$ unconditionally — even for failed attempts (morphism does not exist). This is the No Free Insight law applied to relational claims.
- `MORPH_TENSOR`$(dst, f, g, \Delta\mu)$: tensor product $f \otimes g$. Requires that the union modules $\mathrm{src}(f) \cup \mathrm{src}(g)$ and $\mathrm{tgt}(f) \cup \mathrm{tgt}(g)$ already exist in the graph.
- `MORPH_GET`$(dst, f, sel, \Delta\mu)$: read field from morphism $f$: selector 0 = source, 1 = target, 2 = coupling length, 3 = is_identity.

### 30.1.3 Category Laws

**Theorem 30.1** (Relational Composition is Associative (kernel/CategoryLaws.v) — **(S)**)**.** *For any three morphisms $f : A \to B$, $g : B \to C$, $h : C \to D$ with composable endpoints:*

$$(f; g); h = f; (g; h)$$

*in the relational sense (coupling pairs of the composite). Zero Admitted.*

**Theorem 30.2** (Graph Composition Satisfies Laws (kernel/CategoryBridge.v) — **(S)**)**.** *The operation `graph_compose_morphisms` satisfies associativity and unitality. Additionally:*

`categorical_extension_nofi_consistent`

*MORPH and COMPOSE can carry $\Delta\mu = 0$ and are not in the mandatory-cost class. The NoFI policy enters through `MORPH_ASSERT`: asserting that a built morphism chain exists and activating the cert channel always costs $S(\Delta\mu) \geq 1$. You can build the chain for free; you cannot certify it for free.*

**Theorem 30.3** (Tensor Bifunctoriality (kernel/CategoryMonoidal.v) — **(S)**)**.** *`graph_tensor_morphisms` is a bifunctor: it satisfies the interchange law*

$$(f \otimes g); (h \otimes k) = (f; h) \otimes (g; k)$$

*for composable, disjoint pairs.*

### 30.1.4 The Categorical Separation Theorem

**Theorem 30.4** (Categorical Separation (kernel/PartitionSeparation.v, §10) — **(S)**)**.** *There exist states $s_1, s_2$ such that:*

$$s_1.\texttt{vm\_regs} = s_2.\texttt{vm\_regs} \wedge s_1.\texttt{vm\_mem} = s_2.\texttt{vm\_mem} \wedge s_1.\texttt{vm\_mu} = s_2.\texttt{vm\_mu} \wedge s_1.\texttt{vm\_err} = s_2.\texttt{vm\_err}$$

*but $s_1.\texttt{vm\_graph.pg\_morphisms} \neq s_2.\texttt{vm\_graph.pg\_morphisms}$.*

*Proof.* Construct $s_1$ with `pg_morphisms` $= [(0, ms)]$ for any morphism state $ms$, and $s_2$ with `pg_morphisms` $= []$. Arrange identical `vm_regs`, `vm_mem`, `vm_mu`, `vm_err` by choosing costs and instruction sequences accordingly. Distinction by `discriminate`. □ □

**Remark 30.1** (Significance)**.** A classical machine (exposing only registers, memory, and a cost counter) cannot distinguish $s_1$ from $s_2$. The Thiele Machine distinguishes them in one probe instruction: `MORPH_DELETE` 0 succeeds on $s_1$ and errors on $s_2$. The categorical layer records *how* knowledge was derived, not just *what* the final values are.

This is the machine's claim to being a *knowledge auditor*: any party claiming to have verified a structural relation must carry a $\mu$-receipt $\geq$ the minimum construction cost. The receipt is unforgeable by the NoFI theorem applied to `MORPH_ASSERT`.

## 30.2 The Conservation as Functor (Archived)

**Theorem 30.5** (Conservation as Functor (Universe.v — reference-only))**.** *Define two categories:*

- $\mathbf{C}_{\text{phys}}$*: Objects = universe states (particle momenta lists). Morphisms = paths of momentum-conserving interactions.*
- $\mathbf{C}_{\text{logic}}$*: Objects = total momentum values. Morphisms = equality proofs.*

*The functor $F : \mathbf{C}_{\text{phys}} \to \mathbf{C}_{\text{logic}}$ maps $F(s) = \sum s$ (total momentum). Then:*

$$\text{Path}(s_1, s_2) \implies \sum s_1 = \sum s_2$$

*Conservation of momentum emerges as the functorial image. Observation is a structure-preserving map from physics to logic.*

# Chapter 31

# Audit Infrastructure

**Theorem 31.1** (CatNet Integrity (CatNet.v — reference-only))**.** *Adding a new entry to a valid hash-chain audit log produces a valid chain. If the logic oracle detects inconsistency, execution halts and no further entries are written (paradox halting).*

# Chapter 32

# Falsifiable Predictions

## 32.1 Linear Scaling of Structural Cost

**Axiom 32.1** (Linear Scaling (kernel/FalsifiablePrediction.v)). The $\mu$-cost of maintaining coherence of $N$ entangled qubits scales as $O(N)$ per time step.

**Prediction**: Large-scale quantum computers will encounter a fundamental (not merely technical) decoherence noise floor proportional to entanglement complexity.

## 32.2 CHSH Regime Separation

**Prediction**: Experiments probing the boundary between quantum ($|S| \leq 2\sqrt{2}$) and supra-quantum ($|S| > 2\sqrt{2}$) correlations should find that exceeding the Tsirelson bound requires measurably higher energy dissipation, scaling as:

$$\Delta E \geq k_B T \ln 2 \cdot (|S| - 2\sqrt{2})$$

## 32.3 Metric Deformation

Since $d_\mu$ depends on information content, regions of high structural complexity effectively expand the metric.

**Prediction**: In high-complexity computations, effective signal latency will increase relative to vacuum propagation. (This prediction relies on interpreting $d_\mu$ as physically meaningful, which is currently an analogy rather than a derived result.)

## 32.4 Architectural Permanence

**Theorem 32.1** (Optimal Quartet (ArchTheorem.v — reference-only)). *The four partition discovery strategies (Louvain, Spectral, Degree, Balanced) achieve classification accuracy $> 90\%$ across all problem classes. No alternative configuration exceeds this quartet's accuracy. The configuration is architecturally final.*

## 32.5 Coupling Constant Prediction

**Definition 32.1** (Thiele $\alpha$ Limit (thiele_manifold/PhysicalConstants.v)). The asymptotic density of self-referential programs in the $n$-bit state space:

$$\alpha = \lim_{n\to\infty} \frac{A_{\text{interaction}}(n)}{V_{\text{spacetime}}(n)} = \lim_{n\to\infty} \frac{n+1}{2^n}$$

This converges to 0, but the *non-asymptotic* structure at physical scales should relate to coupling constants.

# Appendix A

# Proof File Index

The table below maps each theorem to its Coq source file. Entries marked — *reference-only* are theoretical results that belong to an extended corpus; the corresponding Coq files are not present in the active kernel tree and their proofs are not machine-checked in the current build. Everything unmarked lives in the active corpus and compiles clean.

| **Result** | **Coq Source** |
|---|---|
| No Free Insight (Thm. 5.1) | `nofi/NoFreeInsight_Theorem.v` |
| $\mu$-Chaitin | `nofi/MuChaitinTheory_Theorem.v` |
| Cost = Complexity | `theory/CostIsComplexity.v` |
| No Free Lunch | `theory/NoFreeLunch.v` |
| No Arbitrage | `kernel/NoArbitrage.v` |
| TM Embedding | `thielemachine/coqproofs/Embedding_TM.v` |
| TM $\rightarrow$ Minsky | `modular_proofs/TM_to_Minsky.v` |
| Strict Subsumption | `thielemachine/coqproofs/Subsumption.v` |
| Semantic Strictness | `thielemachine/coqproofs/ThieleFoundations.v` |
| Confluence | `thielemachine/coqproofs/Confluence.v` |
| Tensoriality | `thielemachine/coqproofs/ThieleUnificationTensor.v` |
| $\mu$-Conservation | `kernel/MuLedgerConservation.v` |
| Receipt Integrity | `kernel/ReceiptIntegrity.v` |
| Noether / Gauge | `kernel/KernelNoether.v` |
| Gauge Trace Preservation | `thielemachine/coqproofs/RepresentationTheorem.v` |
| 2D Necessity | `quantum_derivation/TwoDimensionalNecessity.v` |
| Complex Necessity | `quantum_derivation/ComplexNecessity.v` |
| Born Rule | `kernel/BornRule.v` |
| Born Rule (Tensor) | `kernel/BornRuleFromSymmetry.v` |
| Unitarity | `kernel/Unitarity.v` |
| Purification | `kernel/Purification.v` |
| No-Cloning | `kernel/NoCloning.v` |
| No-Cloning (Machine-Native) | `kernel/NoCloningFromMuMonotonicity.v` |
| Observation Irreversibility | `quantum_derivation/ObservationIrreversibility.v` |
| Collapse Determination | `quantum_derivation/CollapseDetermination.v` |
| Tsirelson Derivation | `kernel/TsirelsonDerivation.v` |
| Tsirelson (Algebraic) | `kernel/TsirelsonFromAlgebra.v` |
| Tsirelson (General) | `kernel/TsirelsonGeneral.v` |
| CHSH Separation | `thielemachine/verification/Deliverable_CHSHSeparati` |
| Emergent Schrödinger | `physics_exploration/EmergentSchrodinger.v` |
| Planck Consistency | `physics_exploration/PlanckDerivation.v` |
| Spacetime Emergence | `kernel/SpacetimeEmergence.v` |
| $\mu$-Geometry | `kernel/MuGeometry.v` |
| 4D Simplicial Complex | `kernel/FourDSimplicialComplex.v` |
| Metric from $\mu$-Costs | `kernel/MetricFromMuCosts.v` |
| Riemann Curvature Tensor | `kernel/RiemannTensor4D.v` |
| Christoffel Symbols | `kernel/RiemannTensor4D.v` |
| Einstein Tensor | `kernel/RiemannTensor4D.v` |
| Stress-Energy Tensor | `kernel/EinsteinEquations4D.v` |
| Curvature from $\mu$-Gradients | `kernel/EinsteinEquations4D.v` |
| Stress-Energy Conservation | `kernel/EinsteinEquations4D.v` |
| Einstein Equations (Vacuum) | `kernel/EinsteinEquations4D.v` |
| Einstein Equations (General) | `kernel/EinsteinEquationsFull.v` |
| Discrete Gaussian Curvature | `kernel/MetricFromMuCosts.v` |
| Information Causality | `kernel/InformationCausality.v` |
| Landauer | `thermodynamic/LandauerDerived.v` |
| Dissipative Embedding | `thielemachine/coqproofs/DissipativeEmbedding.v` |
| Physics Embedding | `thielemachine/coqproofs/PhysicsEmbedding.v` |
| Wave Embedding | `thielemachine/coqproofs/WaveEmbedding.v` |
| Self-Reference | `self_reference/SelfReference.v` |
| Thiele Manifold | `thiele_manifold/ThieleManifold.v` |
| Genesis | `theory/Genesis.v` |
| Full Isomorphism | `kernel/ThreeLayerIsomorphism.v` |
| Curry–Howard–Thiele | `thielemachine/coqproofs/LogicIsomorphism.v` |
| Logic Is Physics | `theory/LogicToPhysics.v` |
| Functor Soundness | `isomorphism/coqproofs/Universe.v` |
| CatNet Integrity | `catnet/coqproofs/CatNet.v` |
| Amortization | `thielemachine/coqproofs/AmortizedAnalysis.v` |
| Optimal Quartet | `theory/ArchTheorem.v` |
| Falsifiable Prediction | `kernel/FalsifiablePrediction.v` |
| Physical Constants | `thiele_manifold/PhysicalConstants.v` |
| Halting Undecidability | `kernel/OracleImpossibility.v` |
| Oracle Cost Bound | `kernel/OracleImpossibility.v` |

# Appendix B

# Epistemological Classification

The following table classifies every major result by its epistemological status.

| Result | Status | Key Assumption |
|---|---|---|
| *Unconditional structural theorems about the machine* | | |
| No Free Insight | **(S)** | Module type contract |
| $\mu$-Chaitin | **(S)** | Module type contract |
| Cost = Complexity | **(S)** | Prefix-free coding |
| No Free Lunch | **(S)** | Faithful representation |
| No Arbitrage $\Rightarrow$ Potential | **(S)** | Additivity + no-arbitrage (toy model) |
| TM Embedding | **(S)** | Standard simulation |
| Strict Subsumption | **(S)** | Partition structure |
| Halting Undecidability | **(S)** | Diagonalization |
| Oracle Cost | **(S)** | Definitional (soundness := cost $\geq 1$) |
| Confluence | **(S)** | Module independence |
| $\mu$-Conservation | **(S)** | Machine semantics only |
| $\mu$-Ledger Necessity | **(S)** | Machine semantics + explicit witnesses |
| $\mu$-Ledger Minimality | **(S)** | Named projection lattice in `NecessityAbstract.v` |
| Receipt Integrity | **(S)** | Hash-chain model |
| Gauge / Noether | **(S)** | Definitional (record structure) |
| Genesis | **(S)** | Definitional isomorphism |
| Three-Layer Isomorphism / comparison contract | **(C)** | Backend decoders satisfy the shared observable projectio |
| Curry–Howard–Thiele | **(S)** | Interpretation map |
| Curvature from $\mu$-Gradients | **(S)** | Lemma 24.5 |
| Vacuum Implies Zero Stress-Energy | **(S)** | Lemma 24.6 |
| Einstein Equations (Vacuum) | **(S)** | Theorem 24.7 |
| Einstein Equations (General full tensor) | **(C)** | Named off-diagonal Ricci premise in `EinsteinEquations` |
| Affine Off-Diagonal Ricci Zero (vertex 1) | **(S)** | Non-uniform isotropic boundary 4-simplex; affine operato |
| Full Tensor EFE at Vertex 1 | **(S)** | `module_structural_mass s 1 = 1` |
| Outer Vertex Characterization | **(S)** | Same affine operator, outer vertices $\{0, 2, 3, 4\}$ |
| Discrete Differential Geometry | **(S)** | Definitions only |
| Partition Ops Free / Certification Non-Free | **(S)** | Machine semantics only |
| CHSH Trial Is Tier 0 | **(S)** | Machine semantics only |
| Three-Tier Observation Taxonomy | **(S)** | Machine semantics only |
| Locally Consistent $\Rightarrow$ Separable Coupling | **(S)** | Coupling definition |
| Local Coupling Bound ($\lvert S\rvert \leq 2$) | **(S)** | Coupling definition |
| $S > 2$ Rules Out Local Couplings | **(S)** | Coupling definition |
| RTL Gap Count = 0 | **(S)** | Machine semantics + hardware spec |
| Classical $P \subseteq \mu P$ | **(S)** | $\mu$-complexity definitions |
| *Conditional derivations (valid given stated axioms)* | | |
| 2D Necessity | **(C)** | Superposition axiom (Axiom 12.1) |
| Complex Necessity | **(C)** | 2D + norm preservation |
| Born Rule Uniqueness | **(C)** | Tensor product consistency (`BornRuleFromSymmetry.v`) |
| Unitarity from $\mu = 0$ | **(C)** | Info conservation (one direction only) |
| No-Cloning | **(C)** | Purity = information |
| Purification | **(C)** | Bloch sphere model |
| Observation Irreversibility | **(C)** | REVEAL semantics |
| Collapse Determination | **(C)** | Entropy = $\log \dim$ |
| Emergent Schrödinger | **(C)** | 2D + antisymmetry |
| Landauer | **(C)** | Second law *assumed* as record field |
| Dissipative Embedding | **(C)** | Irreversibility model |
| Self-Reference | **(C)** | Dimension = complexity |
| CHSH Separation | **(C)** | Numerical witnesses |
| Tsirelson Bound | **(C)** | Algebraic derivation (`TsirelsonFromAlgebra.v`) |
| *Consistency relations (definitions verified, not predictions)* | | |
| Planck's Constant | **(R)** | $\tau_\mu := h/(4E_L)$ |
| Information Causality | **(R)** | Circular: equivalence assumed as hypothesis |
| $\mu$-Geometry | **(R)** | Pseudometric (absolute value on $\mathbb{Z}$) |
| Spacetime Locality | **(R)** | Data isolation in partition graph |
| Born Rule (`BornRule.v`) | **(R)** | $\mu$-consistent hypothesis unused |

**Legend**: **(S)** = unconditional theorem about the machine; **(C)** = valid derivation conditional on stated axiom; **(R)** = internal consistency check.